All cheat sheets

One page-sized reference for each tool, from Linux to Vault.

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Linux

Files, permissions, processes, disks, logs and performance commands you reach for daily.

Navigate and inspect

Command	What it does	
`pwd`	Print the current directory	
`ls -lah`	Long list, hidden files, human sizes	
`cd -`	Return to the previous directory	
`tree -L 2`	Show a directory tree two levels deep	
`stat file`	Size, owner, permissions and timestamps	
`file name`	Detect the type of a file	
`` `du -sh * \ ``	sort -h`	Size of each item here, smallest to largest

Find

```
find /var/log -type f -mtime -1 -size +1M   # modified in the last day, larger than 1 MB
find . -name "*.conf" -not -path "./tmp/*"  # by name, excluding a path
grep -rn "error" /etc/nginx                  # search file contents recursively
type -a python3                              # which executable runs
```

Copy, move, link

```
cp -a src/ dst/              # copy keeping permissions and links
mv old new                   # rename or move
ln -sfn target linkname      # replace a symlink to a directory safely
readlink -f linkname         # resolve the final path
```

Users and permissions

Command	What it does
`id`	Your user, group IDs and groups
`sudo -l`	What you may run with sudo
`chmod 640 file`	Owner read/write, group read, others nothing
`chmod u+x script.sh`	Add execute for the owner
`chown app:app file`	Change owner and group
`umask`	Default permission mask for new files

Permission digits: read = 4, write = 2, execute = 1. Add them per class (owner, group, others), so `755` is `rwxr-xr-x`.

```
useradd -m -s /bin/bash deploy       # create a user with a home directory
usermod -aG docker deploy            # add to a group (keep existing ones)
passwd -l olduser                    # lock an account
getent passwd deploy                 # look up through all name sources
```

Processes and services

```
ps aux --sort=-%mem | head           # top memory users
ps -eo pid,ppid,stat,cmd | grep defunct   # find zombies
top -o %CPU                          # live view sorted by CPU
kill -TERM 1234                      # polite stop
kill -KILL 1234                      # last resort, cannot be caught
pgrep -af nginx                      # find by name with full command

systemctl status nginx               # state and recent log lines
systemctl enable --now nginx         # start now and at boot
systemctl restart nginx
systemctl list-units --failed        # everything that failed
systemctl daemon-reload              # after editing a unit file
```

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Disks and filesystems

```
df -hT                                # space per filesystem with type
df -i                                 # inode usage (full inodes also stop writes)
lsblk -f                              # block devices and filesystems
findmnt /data                         # where and how a path is mounted
du -xh / --max-depth=1 2>/dev/null | sort -h | tail
lsof +L1                              # deleted files still held open
```

Logs

```
journalctl -u nginx --since "1 hour ago"
journalctl -p err -b                  # errors since boot
journalctl -f -u myapp                # follow live
tail -f /var/log/syslog
dmesg -T | tail                       # kernel messages with readable time
```

Performance triage (60 seconds)

```
uptime                                # load averages
vmstat 1 5                            # run queue, swap, I/O wait
free -h                               # memory and swap
iostat -xz 1 3                        # disk utilization and latency
ss -s                                 # socket summary
sar -n DEV 1 3                        # network throughput per interface
```

Symptom	Look at
Load high, CPU idle	I/O wait (`vmstat`, `iostat`)
Memory full, swap busy	`ps aux --sort=-%mem`, OOM messages in `dmesg`
"No space left" but `df -h` fine	`df -i` for inodes
Disk full after deleting	`lsof +L1` for open deleted files
Service flapping	`journalctl -u NAME`, `systemctl status NAME`

Text and data tools

```
grep -rn --include='*.conf' 'listen' /etc/nginx
grep -c 'ERROR' app.log
grep -v '^#' /etc/ssh/sshd_config | sed '/^$/d'
awk -F: '$3 >= 1000 {print $1, $3}' /etc/passwd
awk '{s+=$NF} END {print s}' sizes.txt
sed -n '20,40p' file.txt
sed 's/old/new/g' file.txt
cut -d, -f1,3 data.csv
sort data.txt | uniq -c | sort -rn | head
tr -s ' ' < file.txt
wc -l *.log
diff -u old.conf new.conf
xargs -a hosts.txt -P4 -I{} ping -c1 {}
tee -a out.log < input.txt
```

> **Danger:** `sed -i` *edits the file in place with no undo. Keep a copy first:* `sed -i.bak 's/a/b/' file`. *A mistyped* `xargs` *with* `rm` *can delete the wrong files, so run it with* `echo` *first.*

Archives and transfer

```
tar -czf backup.tar.gz /srv/app
tar -tzf backup.tar.gz | head
tar -xzf backup.tar.gz -C /restore
zip -r site.zip site/
unzip -l site.zip
rsync -avh --dry-run src/ user@host:/dst/
rsync -avh --delete src/ user@host:/dst/
scp file.txt user@host:/tmp/
curl -fsSLO https://example.com/file.tgz
sha256sum file.tgz
```

> **Danger:** `rsync --delete` *removes files at the destination that are missing from the source. Always run it with* `--dry-run` *first, and mind the trailing slash on the source:* `src/` *copies the contents,* `src` *copies the folder.*

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Warning: `tar -xzf` *can overwrite existing files. Extract into an empty folder with* `-C`.

Permissions and ownership, more

```
stat -c '%a %U:%G %n' /etc/passwd
chmod u+x script.sh
chmod -R g+rX /srv/shared
chown -R app:app /srv/app
setfacl -m u:alice:rx /srv/data
getfacl /srv/data
umask
sudo -l
lsattr file
chattr +i /etc/important.conf
```

Warning: `chmod -R 777` *and* `chown -R` *on a wrong path (for example* `/` *or* `/etc`*) can break the system. Check the path with* `ls` *first. Never use 777 as a fix for "permission denied".*

Warning: `chattr +i` *makes a file immutable, even for root. Remember it before you wonder why edits fail.*

Network checks on a host

```
ip -br addr
ip route
ss -tulpn
ss -s
ping -c3 1.1.1.1
dig +short example.com
curl -sS -o /dev/null -w '%{http_code} %{time_total}s\n' https://example.com
traceroute -n example.com
nc -zv host 443
sudo tcpdump -i eth0 -nn port 53 -c 20
```

Warning: `tcpdump` *on a busy host can fill the disk and add load. Always limit it with* `-c`*, a filter and a time limit, and do not capture personal data you are not allowed to keep.*

Cautions for risky commands

Danger: `rm -rf` *has no recycle bin. Never run it with an unquoted variable such as* `rm -rf $DIR/*`*: if* `DIR` *is empty it becomes* `rm -rf /*`*. Use* `rm -rf -- "${DIR:?}"/`.

Warning: `dd of=/dev/sdX` *overwrites a disk with no confirmation. Confirm the device with* `lsblk` *and read the command twice.*

Danger: `mkfs`, `fdisk` *and* `parted` *on a mounted or wrong disk destroy data. Check* `lsblk -f` *and* `findmnt` *first.*

Warning: `kill -9` *gives a process no chance to clean up. Try* `kill` *(SIGTERM) first and wait.*

Warning: `reboot`, `shutdown` *and* `init 0` *over SSH on a remote server can leave you locked out. Confirm that you are on the right host with* `hostname`.

Warning: *Editing* `/etc/fstab`, `/etc/sudoers` *or* `sshd_config` *with a mistake can stop boot or lock you out. Use* `visudo`*, keep a second session open, and test with* `sshd -t` *before you reload.*

Warning: `iptables -F` *and changing the default policy over SSH can cut your own connection. Use a timed rollback such as* `at now + 5 minutes`.

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Git

Everyday commands for commits, branches, remotes, undoing mistakes, tags and recovery.

Set up and first commits

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
git config --global init.defaultBranch main
git config --global pull.ff only          # safe pull default
git init                                  # new repository here
git clone URL                             # copy a repository
git status                                # what is changed, staged, untracked
git add FILE                              # stage a file
git add -p                                # stage pieces of files
git commit -m "Fix retry loop in backup.sh"
git commit --amend --no-edit              # add staged changes to the last commit (unpushed)
```

Area	Meaning
Working tree	The files you edit
Staging area (index)	What goes into the next commit
Repository	The saved history in `.git`

Warning: `git restore FILE` *and* `git reset --hard` *throw away uncommitted edits for good. Run* `git status` *first, and stash if you are unsure.*

Look at history

```
git log --oneline --graph --decorate -15
git log -p -- FILE                        # changes to one file
git log -S"text"                          # commits that added or removed text
git log A..B                              # commits in B not in A
git show HASH                             # one commit in full
git diff                                  # working tree vs staging
git diff --staged                         # staging vs last commit
git diff main...feature                   # what the branch adds
git blame -L 40,60 FILE                   # who last changed each line
```

Branches and merging

```
git branch                                # list branches
git switch -c feature/retry               # create and switch
git switch main
git merge feature/retry                   # merge into current branch
git merge --no-ff feature/retry           # always create a merge commit
git merge --abort                         # give up a merge
git rebase origin/main                    # replay my commits on the latest main
git rebase -i HEAD~4                      # tidy the last four commits
git rebase --abort
git branch -d feature/retry               # delete a merged branch
git cherry-pick HASH                      # copy one commit here
```

Warning: *Never rebase or amend commits that others have already pulled. It creates new commits, and their copies no longer match.*

Remotes

```
git remote -v
git fetch origin                          # download, change nothing local
git pull --ff-only                        # fetch and fast-forward only
git pull --rebase                         # fetch and replay my commits
git push -u origin feature/retry          # first push, set upstream
git push --force-with-lease               # safer forced update (own branches only)
git branch -vv                            # ahead or behind
```

Warning: *Plain* `git push --force` *can erase a colleague's commits. Use* `--force-with-lease` *on your own branch, and block force pushes on shared branches.*

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Undo

Goal	Command
Discard edits to a file	`git restore FILE`
Unstage a file	`git restore --staged FILE`
Fix last commit message	`git commit --amend`
Reverse a shared commit	`git revert HASH`
Undo commit, keep changes staged	`git reset --soft HEAD~1`
Match the remote exactly (loses local work)	`git reset --hard origin/main`
Park work	`git stash push -m "note"` then `git stash pop`
Preview deleting untracked files	`git clean -n`

Tags and releases

```
git tag -a v1.4.0 -m "Release 1.4.0"      # annotated tag
git push origin v1.4.0
git describe --tags                       # v1.4.0-12-g3f9c2ab
git log v1.3.0..v1.4.0 --oneline          # what is in the release
git tag --contains HASH                   # which releases hold this fix
```

Semantic versioning: `MAJOR.MINOR.PATCH`. Breaking change raises MAJOR, new compatible feature raises MINOR, bug fix raises PATCH.

Recover and investigate

```
git reflog                                # where HEAD has been
git branch rescue HASH                    # name a lost commit
git bisect start; git bisect bad; git bisect good v1.3.0
git bisect run ./test.sh                  # automatic search for the first bad commit
git fsck --lost-found                     # unreachable objects
git rm --cached FILE                      # stop tracking, keep the file
```

Big repositories

```
git clone --depth 1 URL                   # latest commit only
git clone --filter=blob:none URL          # fetch file contents on demand
git sparse-checkout set DIR1 DIR2         # check out only these folders
git worktree add ../hotfix hotfix/1.4     # second working folder
git submodule update --init --recursive
git count-objects -vH
```

Warning: *A secret that was committed is compromised, even after you delete it. Revoke it first, then clean history, because old clones keep a copy.*

Safe habits

- Run `git status` before every commit, reset or checkout.
- Keep commits small, with a message that says why.
- Never rewrite commits that others have pulled.
- Never commit secrets. If one is committed, revoke it first.
- Protect `main`: pull requests, review, required checks, no force push.

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Bash scripting

Safe script headers, variables, conditionals, loops, functions and text-processing one-liners.

Safe header

```
#!/usr/bin/env bash
set -euo pipefail
IFS=$'\n\t'
```

Option	Effect
`-e`	Exit when a command fails
`-u`	Treat unset variables as errors
`-o pipefail`	A pipeline fails if any stage fails

Variables and quoting

```
name="web"
echo "Hello, $name"              # double quotes expand variables
echo 'Literal $name'             # single quotes do not
echo "${name:-default}"          # default if unset or empty
echo "${#name}"                  # length
echo "${path##*/}"               # strip longest prefix up to /
echo "${file%.txt}"              # strip suffix
readonly LIMIT=10
```

Always quote expansions: `"$file"`, `"$@"`.

Conditionals

```
if [[ -f "$file" && -r "$file" ]]; then
  echo "readable file"
elif [[ -d "$file" ]]; then
  echo "directory"
else
  echo "missing"
fi
```

Test	True when
`-f F` / `-d F`	F is a file / directory
`-s F`	F exists and is not empty
`-z S` / `-n S`	String is empty / not empty
`A == B`	Strings equal (inside `[[ ]]`)
`N -gt M`	Number greater than

Loops

```
for f in *.log; do
  [[ -e "$f" ]] || continue
  gzip "$f"
done

while IFS= read -r line; do
  echo "line: $line"
done < input.txt

for i in {1..5}; do echo "$i"; done
```

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Functions and arguments

```
usage() { echo "usage: $0 ENV" >&2; exit 2; }

main() {
  local env="${1:-}"
  [[ -n "$env" ]] || usage
  echo "deploying to $env"
}

main "$@"
```

Variable	Meaning
$0	Script name
$1 ... $9	Positional arguments
$@	All arguments, each separately quoted
$#	Number of arguments
$?	Exit status of the last command

Cleanup and temp files

```
tmp="$(mktemp -d)"
trap 'rm -rf "$tmp"' EXIT
```

Text processing

```
grep -E "ERROR|WARN" app.log | wc -l
awk '{print $1}' access.log | sort | uniq -c | sort -rn | head
awk -F, '$3 > 100 {print $1, $3}' data.csv
sed -i.bak 's/old/new/g' config.txt
cut -d: -f1 /etc/passwd
jq -r '.items[].metadata.name' pods.json
xargs -P4 -n1 gzip < filelist.txt
```

Handy patterns

```
command -v jq >/dev/null || { echo "jq required" >&2; exit 1; }
retry() { local n=0; until "$@"; do n=$((n+1)); [[ $n -ge 3 ]] && return 1; sleep $((2**n)); done; \
    }
exec 9>/tmp/job.lock && flock -n 9 || { echo "already running"; exit 1; }
```

Safer scripts

```
#!/usr/bin/env bash
set -Eeuo pipefail
IFS=$'\n\t'
trap 'echo "failed at line $LINENO" >&2' ERR
trap 'rm -rf -- "$tmp"' EXIT
tmp=$(mktemp -d)
: "${ENVIRONMENT:?set ENVIRONMENT}"
command -v jq >/dev/null || { echo "jq is required" >&2; exit 1; }
```

Warning: `set -e` *is not a full safety net. It does not stop on failures inside* `if` *conditions,* `&&` *lists or command substitutions in some cases. Check exit codes of the commands that matter.*

Danger: `rm -rf "$tmp"` *with an empty* `tmp` *is dangerous. The* `:?` *form (*`${tmp:?}`*) stops the script instead.*

Variables and expansion

```
name=${1:-default}
echo "${path##*/}"
echo "${file%.txt}.md"
echo "${#name}"
echo "${name^^}"
arr=(a b c); echo "${arr[@]}" "${#arr[@]}"
declare -A ages=([ann]=30 [bob]=25); echo "${ages[ann]}"
echo "$(date +%F)"
echo $((5 * 3))
printf '%s\n' "${arr[@]}"
```

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Warning: *Always quote variables:* `"$var"`*. An unquoted value with spaces splits into several arguments, and* `*` *is expanded to file names.*

Loops, tests and functions

```
for f in *.log; do [ -e "$f" ] || continue; gzip "$f"; done
while IFS= read -r line; do echo "$line"; done < hosts.txt
[[ -f file && -r file ]] && echo readable
[[ $x =~ ^[0-9]+$ ]] && echo number
case "$1" in start) run ;; stop) halt ;; *) usage ;; esac
retry() { local n=0; until "$@"; do ((++n >= 5)) && return 1; sleep $((2**n)); done; }
```

Input, output and files

```
cmd > out.txt 2> err.txt
cmd > all.txt 2>&1
cmd | tee out.txt
cmd <<'EOT'
literal text, no expansion
EOT
exec 3>>audit.log; echo "ran" >&3
mapfile -t lines < file.txt
find . -name '*.tmp' -print0 | xargs -0 rm -f
flock -n /tmp/job.lock ./job.sh
```

Danger: `>` *truncates the file immediately. A typo such as* `cmd > important.conf` *destroys the content. Use* `set -o noclobber` *or* `>>` *to append.*

Warning: `eval` *runs a string as code. Never pass user input to it. Prefer arrays and* `"${cmd[@]}"`*.*

Danger: `xargs rm` *and* `find -delete` *act at once. Run the command with* `-print` *or* `echo` *first.*

Debugging

```
bash -n script.sh
bash -x script.sh
set -x; risky; set +x
shellcheck script.sh
PS4='+ ${BASH_SOURCE}:${LINENO}: ' bash -x script.sh
echo "${PIPESTATUS[@]}"
```

Warning: `set -x` *prints every command, including ones that contain passwords or tokens. Turn it off around secrets, and do not leave it on in CI logs.*

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Python

Environments, core syntax, files, subprocess, HTTP, JSON, cloud SDK, testing and packaging snippets for automation.

Environments

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install requests==2.32.3
python -m pip freeze > requirements.txt
python -m pip install -r requirements.txt
python -m pip list --outdated
deactivate
```

One virtual environment per project. Call `python -m pip` so pip matches the interpreter.

Core syntax

```
servers = ["web-1", "web-2", "db-1"]
limits = {"cpu": 80, "memory": 90}
web = [s for s in servers if s.startswith("web")]
by_len = {s: len(s) for s in servers}
print(f"{len(web)} web servers, cpu limit {limits.get('cpu', 100)}")

def over(metric: str, value: float) -> bool:
    return value > limits.get(metric, 100)

if __name__ == "__main__":
    raise SystemExit(0)
```

Files and paths

```
from pathlib import Path

p = Path("/etc/myapp/app.conf")
text = p.read_text(encoding="utf-8")
Path("/var/tmp/out").mkdir(parents=True, exist_ok=True)
for f in Path("/var/log/myapp").glob("*.log"):
    print(f.name, f.stat().st_size)
with open("report.txt", "w", encoding="utf-8") as fh:
    fh.write("done\n")
```

Run commands

```
import subprocess

r = subprocess.run(["systemctl", "is-active", "nginx"], capture_output=True, text=True, timeout=10 \
    )
print(r.returncode, r.stdout.strip())
subprocess.run(["tar", "-czf", "/tmp/site.tgz", "/var/www"], check=True)
```

Pass a list, avoid `shell=True`, always set a timeout.

JSON and HTTP

```
import json
import os
import requests

data = json.loads('{"name": "web-1", "cpu": 0.4}')
print(json.dumps(data, indent=2, sort_keys=True))

resp = requests.get(
    "https://api.example.com/v1/servers",
    headers={"Authorization": "Bearer " + os.environ["API_TOKEN"]},
    params={"limit": 50},
    timeout=10,
)
resp.raise_for_status()
items = resp.json()["items"]
```

Always set `timeout`, call `raise_for_status`, read tokens from the environment.

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Errors and logging

```
import logging

logging.basicConfig(level=logging.INFO, format="%(asctime)s %(levelname)s %(message)s")
log = logging.getLogger("job")
try:
    risky()
except ValueError as exc:
    log.exception("failed: %s", exc)
    raise
```

AWS and Azure SDKs

```
import boto3

print(boto3.client("sts").get_caller_identity()["Arn"])
ec2 = boto3.client("ec2", region_name="eu-west-1")
for page in ec2.get_paginator("describe_instances").paginate():
    for res in page["Reservations"]:
        for inst in res["Instances"]:
            print(inst["InstanceId"])

from azure.identity import DefaultAzureCredential
from azure.mgmt.compute import ComputeManagementClient

compute = ComputeManagementClient(DefaultAzureCredential(), "SUBSCRIPTION_ID")
for vm in compute.virtual_machines.list_all():
    print(vm.name)
```

CLI, tests, packaging

```
import argparse

p = argparse.ArgumentParser(prog="tool")
p.add_argument("paths", nargs="+")
p.add_argument("--threshold", type=int, default=85)
args = p.parse_args()
```

```
python -m pip install pytest
python -m pytest -q
python -m pytest -q -k "threshold" -x
python -m pip install .
python -m pip install pipx
```

Mock slow or risky calls with `unittest.mock.patch`; use the `tmp_path` fixture for files.

Environments and packages

```
python3 -m venv .venv && source .venv/bin/activate
python -m pip install -r requirements.txt
python -m pip install --upgrade pip
python -m pip freeze > requirements.lock
python -m pip list --outdated
python -m pip check
python -m pip install -e .
python -m pip install --require-hashes -r requirements.lock
python -m pip_audit
python -V; which python
```

> **Warning:** *Do not run* `sudo pip install`. *It changes system packages and can break the operating system tools. Use a virtual environment.*

> **Warning:** *Installing a package runs its code, and a mistyped name can be a malicious look-alike. Check the name and use a lock file with hashes in production.*

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Debug and profile

```
python -X dev -W error script.py
python -m pdb script.py
python -m trace --trace script.py | head
python -m cProfile -o out.prof -s cumtime script.py
python -m timeit -s 'import json' 'json.dumps({"a": 1})'
python -m http.server 8000 --bind 127.0.0.1
python -m json.tool data.json
python -c 'import sys; print(sys.path)'
```

Warning: `python -m http.server` *serves the current directory to anyone who can reach the port. Bind to* `127.0.0.1`*, and do not run it in a folder with secrets.*

Warning: `breakpoint()` *and* `pdb` *left in code stop a service waiting for input. Remove them before you commit.*

Files, JSON and HTTP in scripts

```
from pathlib import Path
import json, os, subprocess, requests

data = json.loads(Path("config.json").read_text())
Path("out").mkdir(parents=True, exist_ok=True)
r = requests.get(url, timeout=5, headers={"Authorization": f"Bearer {os.environ['TOKEN']}"})
r.raise_for_status()
out = subprocess.run(["kubectl", "get", "pods", "-o", "json"], capture_output=True, text=True, che \
    ck=True, timeout=30)
pods = json.loads(out.stdout)["items"]
```

Warning: `subprocess.run(cmd, shell=True)` *with user input allows command injection. Pass a list of arguments, and keep* `shell=False`*.*

Warning: `requests.get` *without* `timeout` *can hang forever. Always set one, and never disable certificate checks with* `verify=False`*.*

Warning: `pickle.load`*,* `yaml.load` *and* `eval` *on untrusted data can run code. Use* `json` *and* `yaml.safe_load`*.*

Warning: *Do not put tokens or passwords in the code or in logs. Read them from environment variables or a secrets manager.*

Testing and quality

```
python -m pytest -q -x --lf
python -m pytest --cov=app --cov-report=term-missing
python -m ruff check . && python -m ruff format --check .
python -m mypy app
python -m build
python -m twine check dist/*
```

Danger: `twine upload` *publishes to a public index and cannot be undone: a version number can never be reused. Test with the test index first.*

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Networking

Addressing, DNS, HTTP and TLS checks, and the order to troubleshoot connectivity.

Inspect your host

```
ip -br addr                          # interfaces and addresses
ip route                             # routing table
ip route get 10.20.20.15             # which route a destination uses
ss -tulpn                            # listening TCP/UDP ports and owners
ss -s                                # socket summary
```

CIDR quick reference

CIDR	Addresses	Mask
/32	1	255.255.255.255
/30	4	255.255.255.252
/28	16	255.255.255.240
/24	256	255.255.255.0
/20	4,096	255.255.240.0
/16	65,536	255.255.0.0
/8	16,777,216	255.0.0.0

Private ranges: `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`. Cloud providers reserve a few addresses in every subnet.

DNS

```
dig +short example.com A
dig +short example.com MX
dig @8.8.8.8 example.com             # ask a specific resolver
dig +trace example.com               # follow delegation from the root
dig -x 192.0.2.10                    # reverse lookup
getent hosts example.com             # what the system resolver returns
```

Record	Holds
A / AAAA	IPv4 / IPv6 address
CNAME	Alias to another name
MX	Mail servers
TXT	Free text (SPF, verification)
NS	Name servers for the zone

HTTP and TLS

```
curl -sS -o /dev/null -w "%{http_code} %{time_total}s\n" https://example.com
curl -I https://example.com
curl -v --resolve example.com:443:192.0.2.10 https://example.com
openssl s_client -connect example.com:443 -servername example.com </dev/null 2>/dev/null | openssl \
     x509 -noout -dates -subject
```

Status	Meaning
2xx	Success
301 / 302	Redirect (permanent / temporary)
401 / 403	Not authenticated / not allowed
404	Not found
429	Rate limited
502 / 503 / 504	Bad gateway / unavailable / gateway timeout

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Connectivity tests

```
ping -c 3 192.0.2.10                # reachability
traceroute -n 192.0.2.10            # path
mtr -rwc 20 192.0.2.10              # path with loss per hop
nc -zv -w 3 192.0.2.10 5432         # is a TCP port open
tcpdump -ni eth0 port 53 -c 20      # capture a few DNS packets
```

Troubleshooting order

1. Name resolves to the expected address? (`dig`, `getent hosts`)
2. Route exists? (`ip route get`)
3. Host reachable? (`ping`, `mtr`)
4. Port open? (`nc -zv`) - timeout means dropped, refused means nothing listening
5. Firewall rules allow it, in both directions where stateless?
6. TLS valid and trusted? (`openssl s_client`)
7. Application responds correctly? (`curl -v`)

Firewall quick reference

```
iptables -S                         # list rules
nft list ruleset                    # nftables rules
ufw status verbose
firewall-cmd --list-all
```

Name resolution and routing

```
dig example.com A +short
dig @1.1.1.1 example.com MX
dig +trace example.com
dig -x 93.184.216.34 +short
resolvectl status
getent hosts example.com
nslookup example.com
ip route get 8.8.8.8
ip neigh
mtr -rwzc 20 example.com
```

> **Warning:** *DNS answers are cached. After you change a record, old answers live until the TTL expires, so lower the TTL before a planned change, not during it.*

Connectivity and ports

```
nc -zv db.internal 5432
nc -l 8080
curl -v --connect-timeout 5 https://example.com
curl -I https://example.com
curl --resolve example.com:443:10.0.0.10 https://example.com/healthz
openssl s_client -connect example.com:443 -servername example.com </dev/null | openssl x509 -noout \
     -dates -subject
ss -ltnp
ss -tn state established '( dport = :443 )' | wc -l
nmap -p 22,80,443 10.0.0.10
```

> **Warning:** *Scan only hosts and networks that you own or have written permission to test.* `nmap` *against other networks can break rules and trigger security alarms.*

> **Warning:** `curl -k` *and* `--insecure` *turn off certificate checks. Use them only to find a problem, never in scripts.*

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Firewall and packet capture

```
sudo iptables -S
sudo nft list ruleset
sudo ufw status verbose
sudo firewall-cmd --list-all
sudo tcpdump -ni any host 10.0.0.10 and port 443 -c 50 -w /tmp/cap.pcap
sudo tcpdump -nn -r /tmp/cap.pcap | head
ip -s link show eth0
ethtool eth0
```

Warning: *Changing firewall rules over SSH can lock you out. Add an allow rule for your own session first, keep a second session open, and schedule an automatic rollback.*

Danger: *Packet captures can contain passwords, tokens and personal data. Store them securely and delete them when the work is done.*

Cautions for network changes

Warning: *Changing an IP address, route or MTU on a remote host with* `ip addr`, `ip route` *or* `ip link set` *can cut your connection. Use the console or an out-of-band path for risky changes.*

Warning: `ip route add default` *with a wrong gateway breaks all outbound traffic. Check* `ip route` *before and after.*

Warning: *Overlapping CIDR ranges between VPCs, VPNs and clusters cannot be peered later. Plan address space before you create networks.*

Warning: *Disabling reverse path filtering or turning on* `ip_forward` *changes how the host handles traffic. Do it only on purpose and document it.*

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Docker and Compose

Run, build, inspect, clean up and harden containers, plus everyday Compose commands.

Run and manage

```
docker run -d --name web -p 8080:80 nginx:1.27
docker run --rm -it alpine:3.20 sh          # throwaway interactive container
docker ps -a                                 # include stopped
docker stop web && docker rm web
docker exec -it web sh                       # shell in a running container
docker logs -f --tail 50 web
docker cp web:/etc/nginx/nginx.conf ./nginx.conf
```

Flag	Meaning
`-d`	Run in the background
`-p HOST:CONTAINER`	Publish a port
`-e KEY=value`	Set an environment variable
`-v NAME:/path`	Mount a volume
`--network NAME`	Join a network
`--restart unless-stopped`	Restart policy
`--memory 512m --cpus 1`	Resource limits
`--user 10001`	Run as a non-root user

Build and ship

```
docker build -t shop-api:1.8.2 .
docker build --no-cache -t shop-api:dev .
docker tag shop-api:1.8.2 registry.example.com/shop-api:1.8.2
docker push registry.example.com/shop-api:1.8.2
docker pull registry.example.com/shop-api:1.8.2
docker history shop-api:1.8.2
```

Dockerfile essentials

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
USER 10001
EXPOSE 8080
CMD ["python", "app.py"]
```

Rule	Why
Copy dependency files first	Keeps the install layer cached
Pin base image versions	Reproducible builds
Use multi-stage builds	Small final image without build tools
Use exec form for `CMD`	Program is PID 1 and receives signals
Add `.dockerignore`	Smaller context, no secrets copied

Volumes and networks

```
docker volume create pgdata
docker volume ls
docker volume inspect pgdata
docker network create backend
docker network connect backend web
docker network inspect backend
```

On a user-defined network containers resolve each other by name. The default bridge does not.

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Inspect and debug

```
docker inspect web --format '{{.State.ExitCode}} oom={{.State.OOMKilled}}'
docker inspect web --format '{{range .NetworkSettings.Networks}}{{.IPAddress}} {{end}}'
docker stats --no-stream
docker top web
docker events --since 10m
docker run --rm -it --entrypoint sh shop-api:1.8.2
```

Exit code	Meaning
0	Finished normally
1	Application error
125 / 126 / 127	Docker error / not executable / command not found
137	SIGKILL, often out of memory
139	Segmentation fault
143	SIGTERM, normal stop

Clean up

```
docker system df                    # disk used by images, containers, volumes
docker container prune              # remove stopped containers
docker image prune -a               # remove unused images
docker volume prune                 # remove unused volumes (check first)
```

Compose

```
docker compose up -d --build
docker compose ps
docker compose logs -f --tail 50 app
docker compose exec db psql -U app -d appdb
docker compose config               # validate and print the merged file
docker compose down                 # keeps volumes
docker compose down -v              # also deletes volumes: destroys data
```

Hardened run

```
docker run -d --name api \
  --user 10001:10001 --read-only --tmpfs /tmp \
  --cap-drop ALL --security-opt no-new-privileges \
  --memory 512m --cpus 1 \
  registry.example.com/shop-api:1.8.2
```

Avoid `--privileged` and mounting `/var/run/docker.sock` into containers.

Images and registries, more

```
docker image ls --digests
docker history --no-trunc app:1.4
docker image inspect app:1.4 --format '{{.Config.User}} {{.Config.Entrypoint}}'
docker tag app:1.4 registry.example.com/team/app:1.4
docker push registry.example.com/team/app:1.4
docker save -o app.tar app:1.4
docker load -i app.tar
docker buildx build --platform linux/amd64,linux/arm64 -t app:1.4 .
docker scout cves app:1.4
```

> **Warning:** *Do not tag production deployments with* `latest`*. It moves, so two hosts can run different code. Pin a version tag or, better, a digest.*

> **Warning:** *Never copy secrets into an image with* `COPY` *or* `ARG`*. They stay in the image layers and in* `docker history`*. Use build secrets (*`--secret`*) or runtime injection.*

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Run and inspect containers, more

```
docker run -d --name web --restart unless-stopped -p 8080:80 nginx:1.27
docker run --rm -it --user 1000:1000 --read-only --cap-drop ALL alpine sh
docker run --memory 512m --cpus 1.5 app:1.4
docker exec -it web sh
docker logs -f --tail 100 --since 10m web
docker inspect web --format '{{.State.Health.Status}} {{.NetworkSettings.IPAddress}}'
docker stats --no-stream
docker top web
docker diff web
docker cp web:/etc/nginx/nginx.conf ./nginx.conf
docker port web
docker events --since 10m
```

Warning: `--privileged` *gives a container nearly full access to the host. Do not use it unless you understand why, and never mount* `/var/run/docker.sock` *into an untrusted container: it is root on the host.*

Warning: *Publishing with* `-p 8080:80` *listens on all interfaces and Docker edits firewall rules itself. Bind to localhost with* `-p 127.0.0.1:8080:80` *when the service is private.*

Volumes and networks

```
docker volume ls
docker volume inspect pgdata
docker run -v pgdata:/var/lib/postgresql/data postgres:16
docker run -v "$PWD":/work:ro alpine ls /work
docker network ls
docker network create --driver bridge backend
docker network connect backend web
docker network inspect backend
```

Warning: *Data written inside a container is lost when the container is removed. Keep state in named volumes and back them up.*

Warning: *A bind mount such as* `-v /:/host` *exposes the host filesystem. Mount only what you need, and read-only (*`:ro`*) where possible.*

Compose

```
docker compose config
docker compose up -d --build
docker compose ps
docker compose logs -f api
docker compose exec api sh
docker compose pull && docker compose up -d
docker compose down
```

Danger: `docker compose down -v` *also deletes the named volumes, including your database. Use plain* `down` *unless you really want to wipe the data.*

Cleanup and cautions

```
docker system df
docker container prune
docker image prune
docker builder prune --filter until=168h
```

Danger: `docker system prune -a --volumes` *removes every unused image, stopped container and unused volume. On a shared host that can delete someone else's data. Look at* `docker system df` *and prune one resource type at a time.*

Warning: `docker rm -f` *kills a running container without a graceful stop. Use* `docker stop` *first, which sends SIGTERM and waits.*

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CI/CD

Pipeline stages, a Jenkinsfile skeleton, deployment strategies, and declarative delivery commands.

Terms

Term	Meaning
Continuous integration	Merge often; build and test every change automatically
Continuous delivery	Every passing change is releasable; a human chooses when
Continuous deployment	Every passing change is released automatically

Pipeline stages

1. Checkout the exact revision
2. Build the artifact once
3. Test: lint and unit tests first, then integration
4. Analyze: dependency, secret and image scans
5. Publish with an immutable tag
6. Deploy to a test environment and run smoke tests
7. Promote the same artifact to production and verify

Build once, deploy many. Order checks by speed.

Jenkinsfile skeleton

```
pipeline {
  agent { label 'docker' }
  options {
    timeout(time: 30, unit: 'MINUTES')
    disableConcurrentBuilds()
  }
  environment {
    IMAGE = "registry.example.com/shop-api"
    TAG   = "build-${env.BUILD_NUMBER}"
  }
  stages {
    stage('Test') {
      parallel {
        stage('Unit') { steps { sh 'make test' } }
        stage('Lint') { steps { sh 'make lint' } }
      }
    }
    stage('Build') {
      steps { sh 'docker build -t "$IMAGE:$TAG" .' }
    }
    stage('Push') {
      when { branch 'main' }
      steps {
        withCredentials([usernamePassword(credentialsId: 'registry-creds',
                                          usernameVariable: 'U', passwordVariable: 'P')]) {
          sh 'echo "$P" | docker login registry.example.com -u "$U" --password-stdin'
          sh 'docker push "$IMAGE:$TAG"'
        }
      }
    }
  }
  post {
    always { echo 'done' }
  }
}
```

Use single quotes around shell commands that reference secrets, so the shell expands them and Groovy does not.

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Deployment strategies

Strategy	Strength	Weakness
Recreate	Simple	Downtime
Rolling	No downtime, no extra capacity	Slower rollback, mixed versions
Blue-green	Instant switch and rollback	Double capacity
Canary	Limits blast radius	Needs traffic control and metrics

Feature flags separate deploy from release. Remove them when done.

Database changes

1. Expand: add new schema compatibly
2. Migrate: write both, read new, backfill
3. Contract: drop the old schema in a later release

Verification and rollback

```
kubectl rollout status deployment/shop-api -n shop --timeout=180s
kubectl rollout undo deployment/shop-api -n shop
curl -fsS https://shop.example.com/healthz/ready
```

Delivery metrics

Metric	Question
Deployment frequency	How often do we release?
Lead time for changes	Commit to production?
Change failure rate	Share of releases causing problems?
Time to restore	How fast do we recover?

Declarative delivery

```
argocd app list
argocd app diff shop-api-prod
argocd app sync shop-api-prod
argocd app history shop-api-prod
```

Setting	Effect
`automated.prune`	Delete objects removed from configuration
`automated.selfHeal`	Revert manual cluster changes

Pull model: credentials stay in the cluster, drift is detected, and rollback is a configuration revert.

Pipeline security

- Secrets come from a secret store, never from files in the source
- Short-lived credentials via identity federation
- Separate plan or build permissions from deploy permissions
- Scan images and dependencies; fail on high severity
- Deploy only from the main branch after review

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Pipeline hygiene commands

```
yamllint .ci/pipeline.yml
actionlint
docker run --rm -v "$PWD":/src hadolint/hadolint hadolint /src/Dockerfile
shellcheck ci/*.sh
trivy image --severity HIGH,CRITICAL --exit-code 1 app:1.4
trivy fs --scanners vuln,secret .
syft app:1.4 -o spdx-json > sbom.json
cosign sign --key cosign.key registry.example.com/app@sha256:ABC123
cosign verify --key cosign.pub registry.example.com/app@sha256:ABC123
```

Warning: *Do not hide a failing scan with* `|| true` *or* `--exit-code 0` *to make the build green. If a finding is accepted, record it with an owner and an expiry date.*

Warning: *Never print secrets in a pipeline log, and do not run* `env` *or* `set -x` *in a step that has secrets. Masking works only for values that the system knows as secrets.*

Build, test and promote

```
make test
pytest -q --maxfail=1 --junitxml=report.xml
npm ci && npm test
docker build --pull --cache-from app:cache -t app:$SHORT_SHA .
docker push registry.example.com/app:$SHORT_SHA
docker buildx imagetools inspect registry.example.com/app:$SHORT_SHA
curl -fsS https://staging.example.com/healthz
curl -fsS -o /dev/null -w '%{http_code}\n' https://staging.example.com/version
```

Warning: *Use* `npm ci` *or a lock file install in CI, not* `npm install`*. Unlocked installs make builds that cannot be repeated.*

Warning: *Build once and promote the same artifact through environments. Rebuilding for production means that you ship something nobody tested.*

Deploy and rollback

```
kubectl set image deploy/api app=registry.example.com/app:$SHORT_SHA -n shop
kubectl rollout status deploy/api -n shop --timeout=180s
kubectl rollout undo deploy/api -n shop
helm upgrade --install api ./chart -n shop --atomic --timeout 5m
curl -fsS https://example.com/healthz || exit 1
```

Warning: *Do not let a pipeline deploy to production with admin credentials that live forever. Use short-lived credentials, a protected environment and a manual approval for risky changes.*

Warning: *A rollback does not undo database changes. Make migrations backward compatible, and ship them before the code that needs them.*

Warning: *Re-running an old pipeline can redeploy an old version over a newer one. Check the commit and the version before you press retry.*

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Kubernetes

kubectl essentials, manifests, troubleshooting order, RBAC and exam-speed tricks.

Setup for speed

```
alias k=kubectl
source <(kubectl completion bash)
export do="--dry-run=client -o yaml"
k config get-contexts
k config use-context CONTEXT
k config set-context --current --namespace=shop
```

Always confirm the context with `k config current-context` before changing anything.

Read cluster state

```
k get nodes -o wide
k get pods -A
k get pods -n shop -o wide --show-labels
k get deploy,svc,ingress,pvc -n shop
k get events -n shop --sort-by=.lastTimestamp
k describe pod NAME -n shop
k explain pod.spec.containers.resources
k api-resources
k top pod -n shop
```

Create and change

```
k create deployment api --image=nginx:1.27 --replicas=3 $do > deploy.yaml
k run tmp --rm -it --image=busybox:1.36 --restart=Never -- sh
k expose deployment api --port=80 --target-port=8080
k scale deployment api --replicas=5
k set image deployment/api nginx=nginx:1.27.1
k apply -f deploy.yaml
k delete -f deploy.yaml
k label pod NAME tier=backend
k annotate pod NAME note=checked
k edit deployment api
```

Rollouts

```
k rollout status deployment/api
k rollout history deployment/api
k rollout undo deployment/api
k rollout undo deployment/api --to-revision=2
k rollout restart deployment/api
```

Debugging order

```
k get pods -n shop                      # status and restarts
k describe pod NAME -n shop             # events at the bottom
k logs NAME -n shop --previous          # logs of the crashed container
k logs NAME -c CONTAINER -n shop        # a specific container
k exec -it NAME -n shop -- sh
k get endpoints SERVICE -n shop         # empty means selector or readiness problem
k debug NAME -it --image=busybox:1.36   # ephemeral debug container
```

Status	Likely cause
`Pending`	No node fits, or an unbound PVC
`ImagePullBackOff`	Wrong image or missing registry credentials
`CrashLoopBackOff`	Application error; read `--previous` logs
`CreateContainerConfigError`	Missing ConfigMap, Secret or key
`OOMKilled`	Memory limit exceeded
Running, `0/1` ready	Readiness probe failing

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Pod template essentials

```
resources:
  requests: { cpu: 100m, memory: 128Mi }
  limits: { memory: 256Mi }
readinessProbe:
  httpGet: { path: /healthz/ready, port: 8080 }
livenessProbe:
  httpGet: { path: /healthz/live, port: 8080 }
securityContext:
  runAsNonRoot: true
  allowPrivilegeEscalation: false
  readOnlyRootFilesystem: true
  capabilities: { drop: ["ALL"] }
```

Services

Type	Reachable from
ClusterIP	Inside the cluster
NodePort	Node IP on a high port
LoadBalancer	Cloud load balancer address
ExternalName	DNS alias for an external name

Service DNS name: `SERVICE.NAMESPACE.svc.cluster.local`.

RBAC

```
k create role reader --verb=get,list,watch --resource=pods -n shop
k create rolebinding rb --role=reader --serviceaccount=shop:app -n shop
k auth can-i list pods -n shop --as=system:serviceaccount:shop:app
k auth can-i --list -n shop
```

Nodes and maintenance

```
k cordon NODE                                  # stop new scheduling
k drain NODE --ignore-daemonsets --delete-emptydir-data
k uncordon NODE
k taint nodes NODE dedicated=gpu:NoSchedule
k label node NODE disktype=ssd
```

Storage

Item	Note
PVC ReadWriteOnce	One node at a time
PVC ReadWriteMany	Needs a file-capable storage class
Reclaim Delete	Claim deletion deletes the disk
StatefulSet	Stable names and a volume per Pod

JSONPath and output

```
k get pods -o jsonpath='{.items[*].metadata.name}'
k get pod NAME -o jsonpath='{.status.containerStatuses[0].lastState}'
k get secret NAME -o jsonpath='{.data.password}' | base64 -d
k get pods -o custom-columns=NAME:.metadata.name,NODE:.spec.nodeName
```

Context and namespaces

```
kubectl config get-contexts
kubectl config current-context
kubectl config use-context staging
kubectl config set-context --current --namespace=shop
kubectl get ns
kubectl api-resources --namespaced=true
kubectl explain deployment.spec.strategy
kubectl auth can-i delete pods -n shop
kubectl auth can-i --list -n shop
```

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Warning: *Check* `kubectl config current-context` *before any change. Most production accidents are the right command on the wrong cluster. Consider a prompt that shows the context.*

Inspect and debug, more

```
kubectl get pods -A -o wide --field-selector status.phase!=Running
kubectl get events -n shop --sort-by=.lastTimestamp
kubectl describe pod api-0 -n shop
kubectl logs api-0 -c app --previous --tail=200
kubectl logs -l app=api --all-containers --since=15m
kubectl exec -it api-0 -c app -- sh
kubectl debug -it api-0 --image=busybox:1.36 --target=app
kubectl debug node/worker-1 -it --image=busybox:1.36
kubectl port-forward svc/api 8080:80 -n shop
kubectl cp shop/api-0:/tmp/dump.txt ./dump.txt
kubectl top pod -n shop --containers
kubectl top node
kubectl get pod api-0 -o yaml | less
```

Warning: `kubectl exec` *and* `debug` *give shell access to a running workload. Do not change files or install tools in a production container: the change is lost and may hide the real cause.*

Warning: `kubectl port-forward` *exposes the service to your machine only, but anything you paste through it goes to production. Never point a test tool at it.*

Change safely

```
kubectl diff -f deploy.yaml
kubectl apply -f deploy.yaml --dry-run=server
kubectl apply -f deploy.yaml
kubectl rollout status deploy/api -n shop
kubectl rollout undo deploy/api -n shop
kubectl scale deploy/api --replicas=5 -n shop
kubectl set image deploy/api app=registry.example.com/api:1.5 -n shop
kubectl annotate deploy/api kubernetes.io/change-cause="bump to 1.5"
kubectl label pod api-0 debug=true
kubectl patch deploy api -p '{"spec":{"replicas":3}}'
```

Warning: `kubectl edit` *and* `patch` *change live objects that are not in your source files, so the next* `apply` *or pipeline run can undo them silently, or the cluster can drift. Prefer changing the manifest and applying it.*

Warning: `kubectl scale --replicas=0` *stops the service. Do it only on purpose and say so in the incident channel.*

Nodes and maintenance, more

```
kubectl get nodes -o wide
kubectl describe node worker-1 | sed -n '/Taints/,/Conditions/p'
kubectl cordon worker-1
kubectl drain worker-1 --ignore-daemonsets --delete-emptydir-data --timeout=300s
kubectl uncordon worker-1
kubectl taint nodes worker-1 dedicated=gpu:NoSchedule
kubectl get pdb -A
```

Danger: `kubectl drain` *evicts pods. With* `--delete-emptydir-data` *the data in emptyDir volumes is lost, and a PodDisruptionBudget of 0 can block it forever. Drain one node at a time and watch the rollout.*

Danger: `kubectl delete node` *removes the node object, not the machine. Drain first and confirm the cloud instance state.*

Secrets and config

```
kubectl create secret generic db --from-literal=password='change-me' -n shop --dry-run=client -o y \
    aml
kubectl get secret db -n shop -o jsonpath='{.data.password}' | base64 -d
kubectl create configmap app-config --from-file=app.properties
kubectl get cm app-config -o yaml
kubectl rollout restart deploy/api -n shop
```

Warning: *Base64 is not encryption. Anyone who can read a Secret can read its value. Restrict access with RBAC, and do not paste decoded values into chat or tickets.*

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Warning: *Passwords on the command line end up in shell history. Prefer files or a secrets manager.*

Cautions for destructive commands

Danger: `kubectl delete -f` *and* `kubectl delete ns shop` *remove everything in the file or namespace, including volumes with* `Delete` *reclaim policy. Run* `kubectl get all -n shop` *and use* `--dry-run=server` *first.*

Danger: `kubectl delete pod --force --grace-period=0` *skips graceful shutdown and can leave a StatefulSet pod running twice. Use it only for stuck terminating pods after you confirm the node is gone.*

Danger: `kubectl delete pvc` *can delete the underlying disk. Check* `kubectl get pv` *for the reclaim policy first.*

Danger: *Commands with* `-A` *or* `--all` *act on every namespace. Read the output of the matching* `get` *before you run* `delete`*.*

Warning: *Do not bind* `cluster-admin` *to users or service accounts for convenience. Grant the smallest Role that works.*

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Terraform

Workflow commands, HCL syntax, state operations, meta-arguments and safe pipeline practices.

Workflow

```
terraform init
terraform fmt -recursive
terraform validate
terraform plan -out=tfplan
terraform apply tfplan
terraform destroy
terraform output
terraform output -raw bucket_name
```

Plan symbol	Meaning
+	Create
~	Update in place
-	Destroy
-/+	Replace (destroy and recreate)

Read the summary line and every `-` or `-/+` before you apply.

Configuration skeleton

```
terraform {
  required_version = ">= 1.6"
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = var.region
}

variable "region" {
  type    = string
  default = "eu-west-1"
}

resource "aws_s3_bucket" "logs" {
  bucket = "acme-logs-${var.environment}"
}

output "bucket_arn" {
  value = aws_s3_bucket.logs.arn
}
```

Variables

```
variable "environment" {
  type        = string
  description = "dev, staging or prod"
  validation {
    condition     = contains(["dev", "staging", "prod"], var.environment)
    error_message = "Must be dev, staging or prod."
  }
}

variable "db_password" {
  type      = string
  sensitive = true
}
```

Precedence (low to high): default, `TF_VAR_name`, `terraform.tfvars`, `-var-file`, `-var`.

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Expressions

```
locals {
  prefix = "shop-${var.environment}"
  is_prod = var.environment == "prod"
  names   = [for n in var.names : upper(n)]
  size    = local.is_prod ? "m5.large" : "t3.micro"
}
```

Function	Example
`merge(a, b)`	Combine maps
`lookup(map, key, default)`	Safe map access
`cidrsubnet(cidr, 8, 1)`	Carve a subnet
`join(",", list)`	Make a string
`try(expr, fallback)`	Fall back on error

Experiment with `terraform console`.

Meta-arguments

```
resource "aws_subnet" "multi" {
  for_each          = var.subnets          # map or set, stable keys
  vpc_id            = aws_vpc.main.id
  cidr_block        = each.value
}

resource "aws_instance" "web" {
  count = var.enabled ? 1 : 0              # on/off switch
  lifecycle {
    prevent_destroy       = true
    create_before_destroy = true
    ignore_changes        = [tags]
  }
}
```

Prefer `for_each` to `count` for collections; `count` shifts indexes when an item is removed.

State

```
terraform state list
terraform state show aws_s3_bucket.logs
terraform state rm aws_s3_bucket.legacy     # forget, do not delete
terraform apply -refresh-only               # record reality without changing it
terraform force-unlock LOCK_ID              # only when no run is active

moved {
  from = aws_s3_bucket.logs
  to   = aws_s3_bucket.audit_logs
}

import {
  to = aws_s3_bucket.legacy
  id = "acme-legacy-reports"
}
```

Remote backend

```
terraform {
  backend "s3" {
    bucket       = "acme-terraform-state"
    key          = "shop/prod/terraform.tfstate"
    region       = "eu-west-1"
    encrypt      = true
    use_lockfile = true
  }
}
```

Enable bucket versioning, restrict access, and keep one state per environment.

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Modules

```
module "site" {
  source  = "../../modules/static-site"
  name    = "marketing-prod"
}

output "site_bucket" {
  value = module.site.bucket_name
}
```

Pin versions of modules you do not own. Run `terraform init` after adding a module.

Pipeline checks

```
terraform fmt -check -recursive
terraform validate
tflint
trivy config .
terraform plan -detailed-exitcode     # 0 none, 1 error, 2 changes
terraform test
```

Workflow with a saved plan

```
terraform fmt -recursive
terraform validate
terraform init -upgrade
terraform plan -out=tfplan -input=false
terraform show tfplan
terraform show -json tfplan | jq '.resource_changes[] | select(.change.actions != ["no-op"]) | {a: \
     .address, c: .change.actions}'
terraform apply tfplan
terraform output -json
terraform providers
terraform version
```

Danger: *Read the plan before you apply it, especially lines marked* `destroy` *or* `must be replaced`*. A change to one argument can force a new resource, such as a database.*

Warning: `terraform apply -auto-approve` *skips the human check. Use it only in a pipeline that applies a reviewed, saved plan.*

State inspection and surgery

```
terraform state list
terraform state show aws_instance.web
terraform state pull > backup.tfstate
terraform state mv aws_instance.web aws_instance.app
terraform state rm aws_instance.old
terraform import aws_s3_bucket.logs acme-logs
terraform force-unlock LOCK_ID
terraform refresh
terraform taint aws_instance.web
terraform apply -replace=aws_instance.web
```

Danger: *Always run* `terraform state pull > backup.tfstate` *before* `state mv`, `state rm` *or* `import`*. A wrong state edit can make Terraform destroy or duplicate real resources.*

Danger: `terraform force-unlock` *is only safe when you are sure that no other run is active. Check who holds the lock first.*

Warning: *Do not edit the state file by hand and do not commit it to a shared repository: it can contain secrets in plain text. Use a remote backend with encryption and locking.*

Warning: `terraform taint` *is deprecated. Use* `apply -replace=ADDRESS`*, which shows the effect in the plan.*

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Targeting, variables and workspaces

```
terraform plan -target=module.network
terraform plan -var 'env=staging' -var-file=staging.tfvars
terraform workspace list
terraform workspace new staging
terraform workspace select prod
terraform console
export TF_LOG=INFO TF_LOG_PATH=tf.log
TF_VAR_db_password="$DB_PASSWORD" terraform plan
```

Warning: `-target` *skips dependencies and leaves the state out of step with the code. Use it only for recovery, then run a full plan.*

Warning: `TF_LOG=TRACE` *can write secrets and tokens into the log. Do not attach such logs to tickets.*

Warning: *Check* `terraform workspace show` *before apply. Applying to the wrong workspace is a common cause of production changes.*

Destroy

```
terraform plan -destroy -out=destroy.tfplan
terraform show destroy.tfplan | less
terraform destroy -target=aws_instance.test
lifecycle { prevent_destroy = true }
```

Danger: `terraform destroy` *deletes everything in the state, including databases and buckets. Always review* `plan -destroy`*, set* `prevent_destroy` *on critical resources, and turn on deletion protection in the cloud.*

Warning: *Provider upgrades with* `init -upgrade` *can change behaviour. Pin versions in* `required_providers` *and read the changelog.*

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SRE practices

SLO maths, error budget policy, incident roles and updates, postmortem template, toil and reliability patterns.

Definitions

Term	Meaning
SLI	A measurement: good events divided by valid events
SLO	Internal target for an SLI over a window
SLA	Contract with consequences; looser than the SLO
Error budget	1 minus the SLO

Downtime allowed per 30 days

SLO	Allowed downtime
99%	about 7 h 12 min
99.9%	about 43 min
99.95%	about 21 min
99.99%	about 4 min

A 30-day window is 43,200 minutes. Multiply by the error budget fraction.

Error budget policy

Budget state	Action
Healthy	Ship features, run experiments
Under 25% left	Stricter reviews, smaller releases
Exhausted	Pause risky releases; prioritize reliability work

Agree the policy before you need it. Never using the budget also means the target may be too strict.

Burn rate

Burn rate	Meaning	Typical action
1	Budget lasts exactly the window	Normal
6	Budget gone in about 5 days	Ticket
14	Budget gone in about 2 days	Page

Alert on a long and short window together so you are paged for sustained burn only.

Incident roles

Role	Job
Incident commander	Coordinates and decides; does not debug
Operations lead	Investigates and changes systems
Communications lead	Posts updates on a schedule
Scribe	Keeps the timeline

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Severity

Level	Impact	Updates
SEV1	Major outage or data loss	Every 15 to 30 minutes
SEV2	Significant degradation	Every 30 to 60 minutes
SEV3	Minor, workaround exists	Working hours

Incident sequence

1. Detect and acknowledge
2. Triage: impact, scope, severity, open a channel
3. Mitigate first (roll back, fail over, disable a flag)
4. Diagnose with evidence; ask what changed
5. Resolve and confirm recovery holds
6. Schedule the postmortem

One change at a time, announced before it is made.

Status update template

```
HH:MM UTC - SEVn - Short title
Impact: who is affected and how
Status: what we know and what we are doing
Next update: HH:MM UTC
```

Postmortem template

Section	Content
Summary	What broke, how long, who was affected
Impact	Users, requests, budget consumed
Timeline	Timestamped events
Contributing factors	The chain of conditions
Went well / badly / lucky	Honest reflection
Actions	Specific, owned, dated: prevent, detect, mitigate

Blameless: ask how the system made the mistake easy, not who made it.

Toil

Toil is manual, repetitive, automatable, reactive work with no lasting value that grows with the service.

Reduce by	Example
Automating	Script or controller replaces the procedure
Removing the cause	Fix the bug instead of restarting
Self-service	Developers do it without a ticket
Stopping	Drop work that adds no value

Keep toil below about half of an engineer's time.

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Reliability patterns

Pattern	Rule
Timeouts	Every network call; shorter than the caller's deadline
Retries	Idempotent only, exponential backoff with jitter, two or three attempts, retry at one layer
Circuit breaker	Fail fast when a dependency is failing; closed, open, half-open
Bulkhead	Separate pools per dependency
Graceful degradation	Optional features fail without breaking the critical path
Load shedding	Reject excess requests quickly (429 or 503 with Retry-After)
Rate limits	Per-client so one caller cannot harm others

Game day checklist

- Hypothesis written down
- Blast radius limited and agreed
- Abort condition defined
- Quiet period, people watching
- Findings turned into tracked actions

SLO and error budget maths

```
sum(rate(http_requests_total{status!~"5.."}[30d])) / sum(rate(http_requests_total[30d]))
1 - (sum(rate(http_requests_total{status!~"5.."}[1h])) / sum(rate(http_requests_total[1h])))
(sum(rate(http_requests_total{status=~"5.."}[1h])) / sum(rate(http_requests_total[1h]))) / 0.001

99.9 percent over 30 days   = 43.2 minutes of error budget
99.95 percent over 30 days  = 21.6 minutes
99.99 percent over 30 days  = 4.3 minutes
burn rate 14.4 for 1 hour   = 2 percent of the monthly budget gone
burn rate 6 for 6 hours     = 5 percent of the monthly budget gone
```

Warning: *An SLO of 100 percent is not a goal: it leaves no room for change. Pick a target that users notice, and write down what happens when the budget is gone.*

Warning: *Measure the SLI where users are (load balancer, synthetic probe), not only inside the service. A healthy process behind a broken proxy still looks fine from the inside.*

Incident commands and habits

```
date -u
kubectl rollout history deploy/api -n shop
kubectl rollout undo deploy/api -n shop
kubectl get events -A --sort-by=.lastTimestamp | tail -30
curl -s -o /dev/null -w '%{http_code} %{time_total}\n' https://example.com/healthz
journalctl -u api --since "30 min ago" -p warning

1. Declare, name an incident commander, open one channel.
2. Post the first update within 10 minutes, then at a fixed interval.
3. Mitigate first (roll back, fail over, turn a flag off), find the cause later.
4. Write down every change with a UTC time.
5. Confirm recovery on the user-facing graph, then close.
6. Hold a blameless review within 5 working days. Track actions with an owner and a date.
```

Warning: *In an incident do not make several changes at once. You will not know which one helped, or which one made it worse.*

Warning: *Do not restart a failing service just to see: you can lose the evidence in memory. Capture logs,* `dmesg` *and thread or heap information first when it is safe.*

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Warning: *Never paste secrets or personal data into an incident channel or ticket. It is kept for a long time and many people can read it.*

Capacity and resilience checks

```
kubectl top nodes
kubectl get hpa -A
kubectl get pdb -A
df -h; df -i
free -m; vmstat 1 5
iostat -xz 1 5
curl -s localhost:9090/api/v1/query --data-urlencode 'query=predict_linear(node_filesystem_avail_b \
    ytes[6h], 7*86400)' | jq .
```

Warning: *Load and chaos tests belong in a test environment first. In production, start small, have a way to stop, have a named owner and tell the on-call team beforehand.*

Warning: *Do not remove an alert because it is noisy. Fix the cause or the threshold, and keep the page for something a human must do now.*

Warning: *Untested backups are not backups. Restore into a clean environment on a schedule and measure how long it takes.*

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DevSecOps and security

Scanning, secrets, signing, container and cluster hardening, cloud identity checks and incident first steps.

Scan code, dependencies and images

```
trivy fs --severity HIGH,CRITICAL --exit-code 1 .
trivy image --ignore-unfixed --severity HIGH,CRITICAL registry.example.com/app:1.4
trivy config ./terraform
checkov -d ./terraform
semgrep scan --config auto .
syft registry.example.com/app:1.4 -o spdx-json > sbom.json
grype sbom:./sbom.json --fail-on high
```

> **Caution:** *A clean scan means no known issues in the packages the tool recognised. It does not prove the application is safe.*

Secrets

```
trufflehog filesystem . --only-verified
detect-secrets scan > .secrets.baseline
aws sts get-caller-identity
aws iam list-access-keys --user-name deploy
aws iam update-access-key --user-name deploy --access-key-id AKIA... --status Inactive
```

> **Warning:** *Disable or rotate a leaked key first. Removing it from the repository does not make it safe.*

Sign and verify artifacts

```
cosign generate-key-pair
cosign sign --key cosign.key registry.example.com/app@sha256:abc123
cosign verify --key cosign.pub registry.example.com/app@sha256:abc123
cosign attest --predicate sbom.json --type spdxjson --key cosign.key registry.example.com/app@sha2 \
    56:abc123
```

Sign by digest, not by tag. A tag can move.

Harden containers

```
docker run --rm --read-only --tmpfs /tmp --cap-drop ALL \
  --security-opt no-new-privileges --pids-limit 200 --memory 256m app:1.4
docker run --rm --user 65532:65532 app:1.4
docker scout cves app:1.4
```

> **Danger:** *Do not use* `--privileged` *or mount* `/var/run/docker.sock` *into application containers. Both give control of the host.*

Kubernetes security checks

```
kubectl auth can-i --list -n shop --as system:serviceaccount:shop:web
kubectl get clusterrolebindings -o wide | grep cluster-admin
kubectl get pods -A -o jsonpath='{range .items[*]}{.metadata.namespace}{" "}{.metadata.name}{" "}{ \
    .spec.containers[*].securityContext.privileged}{"\n"}{end}'
kubectl label ns shop pod-security.kubernetes.io/enforce=restricted
kubectl get networkpolicy -A
kubectl get secret db -n shop -o jsonpath='{.data.password}' | base64 -d
```

> **Warning:** *The last command prints a secret to your terminal and shell history. Use it only when needed and clear the output.*

Cloud identity checks (AWS)

```
aws iam get-account-authorization-details > iam.json
aws iam generate-credential-report && aws iam get-credential-report
aws accessanalyzer list-findings --analyzer-arn ARN
aws s3api get-public-access-block --bucket NAME
aws cloudtrail lookup-events --lookup-attributes AttributeKey=EventName,AttributeValue=ConsoleLogi \
    n
aws ec2 describe-security-groups --filters Name=ip-permission.cidr,Values=0.0.0.0/0
```

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TLS and certificates

```
openssl s_client -connect example.com:443 -servername example.com </dev/null | openssl x509 -noout \
     -dates -issuer -subject
openssl x509 -in cert.pem -noout -text | head -30
nmap -sV --script ssl-enum-ciphers -p 443 example.com
```

Caution: *Scan only systems you own or have written permission to test.*

Incident first steps

```
ss -tunap | head -40
last -a | head
sudo journalctl -u sshd --since "1 hour ago" | grep -i "failed\|accepted"
sudo iptables -I INPUT -s 203.0.113.50 -j DROP
sudo tcpdump -i eth0 -w /secure/capture.pcap host 203.0.113.50
```

Danger: *Do not wipe or reboot a suspected compromised host before you have taken a snapshot or memory capture. You will destroy the evidence.*

Identity and tokens

```
echo "$TOKEN" | cut -d. -f2 | base64 -d 2>/dev/null | jq '{iss,aud,exp,scope}'
aws iam list-users --query 'Users[].UserName' --output text
aws iam get-user --user-name svc-old --query 'User.PasswordLastUsed'
```

Warning: *A JWT is signed, not encrypted. Never put secrets in it, and always check signature, issuer, audience and expiry.*

Application and API checks

```
curl -sI https://example.com | grep -iE 'strict-transport|content-security|x-content-type'
curl -s -o /dev/null -w "%{http_code}\n" https://api.example.com/invoices/10043 -H "Authorization: \
     Bearer $TOKEN"
```

Caution: *Test only your own systems. A 200 for another customer's object means object-level authorization is missing.*

Policy as code

```
opa eval -d policy/ -i plan.json "data.terraform.security.deny"
kubectl apply -f policy.yaml && kubectl get policyreport -A
```

Tip: *Start every new policy in audit mode, review violations with owners, then enforce in stages.*

Network and DNS

```
aws ec2 describe-security-groups --filters Name=ip-permission.cidr,Values=0.0.0.0/0 --query 'Secur \
    ityGroups[].GroupId'
dig +short CNAME promo.example.com
dig +short TXT example.com | grep -i spf
```

Caution: *A CNAME that points to a deleted cloud resource can be claimed by someone else. Remove dangling records.*

Backups and recovery

```
aws s3api put-object-lock-configuration --bucket backups --object-lock-configuration '{"ObjectLock \
    Enabled":"Enabled","Rule":{"DefaultRetention":{"Mode":"COMPLIANCE","Days":30}}}'
aws cloudtrail lookup-events --lookup-attributes AttributeKey=EventName,AttributeValue=DeleteSnaps \
    hot --max-results 5
```

Danger: *Backups reachable with production credentials will be deleted first. Keep an immutable copy in a separate account and test restores.*

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Ansible

Inventory, ad-hoc commands, playbook structure, variables, handlers, roles, Vault and safe rollout flags.

Ad-hoc commands

```
ansible all -i inventory.ini -m ping
ansible web -i inventory.ini -m command -a "uptime"
ansible web -i inventory.ini -m apt -a "name=nginx state=present" --become
ansible-inventory -i inventory.ini --graph
ansible-inventory -i inventory.ini --host web1.example.com
```

Flag	Meaning
`-m`	Module to run
`-a`	Module arguments
`--become`	Run with elevated rights
`--limit`	Restrict to some hosts
`--check`	Show what would change

Running playbooks

```
ansible-playbook -i inventory.ini site.yml
ansible-playbook site.yml --syntax-check
ansible-playbook site.yml --check --diff --limit web1.example.com
ansible-playbook site.yml --tags config --skip-tags slow
ansible-playbook site.yml --start-at-task "Render nginx configuration"
ansible-playbook site.yml --list-tasks
```

Playbook skeleton

```
- name: Configure web servers
  hosts: web
  become: true
  vars:
    app_port: 8080
  tasks:
    - name: Install nginx
      ansible.builtin.apt:
        name: nginx
        state: present
    - name: Render configuration
      ansible.builtin.template:
        src: shop.conf.j2
        dest: /etc/nginx/conf.d/shop.conf
        validate: "nginx -t -c /etc/nginx/nginx.conf"
      notify: Reload nginx
  handlers:
    - name: Reload nginx
      ansible.builtin.service:
        name: nginx
        state: reloaded
```

Read the recap: a clean second run shows `changed=0`.

Control flow

```
- name: Only on Debian
  ansible.builtin.apt:
    name: nginx
  when: ansible_facts['os_family'] == "Debian"

- name: Make users
  ansible.builtin.user:
    name: "{{ item }}"
  loop: [alice, bob]
```

Tool	Use
when	Run a task only if true

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Tool	Use
`loop`	Repeat for each item
`block` / `rescue` / `always`	Try, catch, finally
`changed_when: false`	Mark a read-only task as unchanged
`no_log: true`	Hide secrets from output

Roles and collections

```
ansible-galaxy init roles/nginx
ansible-galaxy install -r requirements.yml
ansible-galaxy collection list
```

Pin versions in `requirements.yml`. Put overridable settings in `defaults/main.yml` and prefix variable names with the role name.

Vault

```
ansible-vault create group_vars/prod/vault.yml
ansible-vault edit group_vars/prod/vault.yml
ansible-vault encrypt_string 'value' --name 'vault_db_password'
ansible-playbook site.yml --vault-id prod@prompt
```

Keep the vault password out of the repository.

Safe rollout

```
- hosts: web
  serial: "20%"
  max_fail_percentage: 10
```

Test with `ansible-lint` and Molecule. Use a canary batch and a health check before continuing.

Troubleshooting

Symptom	First step
UNREACHABLE, publickey	`ssh -vvv` with the same user and key
Missing sudo password	Check become and sudoers
Missing variable	`ansible-inventory --host`, add `default` or `assert`
Slow run	`gather_facts: false`, more forks, `profile_tasks`

Warning: `ansible-playbook` *without* `--limit` *or* `serial` *changes every host in the group at once. Test on one host first.*

Danger: *Never commit the vault password next to the encrypted files. It cancels the protection.*

Warning: `ignore_errors: true` *hides failures and lets later tasks run on a broken base. Fix the cause or use* `rescue`*.*

Warning: `shell` *and* `command` *always report* `changed` *unless you add* `creates` *or* `changed_when`*.*

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Databases (PostgreSQL)

SQL basics, operator queries, indexes and plans, locks, backups, replication, pooling and safe migrations.

Connect and explore

```
psql "host=db1.example.com dbname=shop user=readonly"
psql -h db1.example.com -U readonly shop -c "SELECT count(*) FROM orders;"
pg_isready -h db1.example.com -p 5432
```

psql command	Meaning
`\dt`	List tables
`\d orders`	Describe a table
`\x`	Toggle vertical output
`\timing`	Show query time

Safe changes

```
BEGIN;
SELECT count(*) FROM orders WHERE created_at < '2020-01-01';
DELETE FROM orders WHERE created_at < '2020-01-01';
COMMIT;   -- or ROLLBACK;
```

Never run `UPDATE` or `DELETE` without `WHERE`. Use a read-only role for investigation.

Operator queries

```
SELECT pg_size_pretty(pg_database_size(current_database()));
SELECT pid, state, now() - query_start AS running_for, left(query, 60)
FROM pg_stat_activity WHERE state <> 'idle' ORDER BY running_for DESC;
SELECT pid, pg_blocking_pids(pid) FROM pg_stat_activity
WHERE cardinality(pg_blocking_pids(pid)) > 0;
```

Indexes and plans

```
EXPLAIN ANALYZE SELECT * FROM orders WHERE customer_id = 42;
CREATE INDEX CONCURRENTLY idx_orders_customer ON orders (customer_id);
ANALYZE orders;
```

In the plan	Meaning
Seq Scan on a big table	Reads every row, often needs an index
Estimate far from actual	Stale statistics, run `ANALYZE`
Sort spilling to disk	Needs a better index or more `work_mem`

Timeouts

```
ALTER ROLE app SET statement_timeout = '30s';
ALTER ROLE app SET lock_timeout = '5s';
ALTER ROLE app SET idle_in_transaction_session_timeout = '30s';
SELECT pg_cancel_backend(1234);
```

Backups

```
pg_dump --format=custom --file=shop.dump shop
pg_restore --dbname=shop_check --jobs=4 shop.dump
pg_basebackup -h db1.example.com -D /backups/base -X stream -P
```

A backup counts only after a successful restore test. A replica is not a backup.

Replication

```
SELECT client_addr, state, pg_wal_lsn_diff(sent_lsn, replay_lsn) AS bytes_behind
FROM pg_stat_replication;
SELECT slot_name, active FROM pg_replication_slots;
```

Failover needs quorum, fencing, promotion and client redirection.

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Safe migrations

Change	Safer approach
New index	`CREATE INDEX CONCURRENTLY`
NOT NULL	`CHECK ... NOT VALID`, then `VALIDATE CONSTRAINT`
Rename a column	Expand and contract over several deploys
Large backfill	Small batches, watch replica lag
Any ALTER	Set `lock_timeout` first

Redis quick checks

```
redis-cli INFO memory | grep used_memory_human
redis-cli INFO stats | grep -E "keyspace_hits|keyspace_misses"
redis-cli --scan --pattern 'session:*' | head
```

Use `SCAN`, never `KEYS *`, on production. Set `maxmemory` and an eviction policy.

Danger: `DELETE` *or* `UPDATE` *without* `WHERE` *changes every row. Use a transaction and check the row count before you commit.*

Warning: *A plain* `CREATE INDEX` *blocks writes while it builds. Use* `CONCURRENTLY` *on busy tables.*

Danger: *Never delete files inside the data directory or* `pg_wal` *by hand. It can destroy the database.*

Warning: `KEYS *` *on a production Redis blocks every client. Use* `SCAN`.

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Go

Toolchain commands, types, errors, interfaces, goroutines, context, HTTP, testing and build flags for services and CLIs.

Toolchain

```
go version
go mod init example.com/shop
go mod tidy
go build ./...
go run ./cmd/api
go test ./... -race -cover
go vet ./...
gofmt -l .
go install ./cmd/api
```

Warning: *Commit* `go.mod` *and* `go.sum`*. Without* `go.sum`*, builds cannot verify that dependencies are unchanged.*

Types, slices and maps

```
type Order struct {
    ID    string
    Items []Item
    Tags  map[string]string
}

names := make([]string, 0, 8)
names = append(names, "a")
m := map[string]int{"a": 1}
if v, ok := m["b"]; !ok { _ = v }
```

Warning: *Writing to a nil map panics. Create maps with* `make` *or a literal. Reading from a nil map is safe.*

Errors and defer

```
f, err := os.Open(path)
if err != nil {
    return fmt.Errorf("open config %q: %w", path, err)
}
defer f.Close()

if errors.Is(err, os.ErrNotExist) { /* handle */ }
var pe *fs.PathError
if errors.As(err, &pe) { /* inspect */ }
```

Warning: `defer` *in a loop runs when the function returns, not each iteration. Move the body into a function to release resources early.*

Interfaces

```
type Store interface {
    Get(ctx context.Context, id string) (Order, error)
}

func NewService(s Store) *Service { return &Service{store: s} }
```

Accept interfaces, return concrete types. Keep interfaces small and define them where they are used.

Goroutines and channels

```
var wg sync.WaitGroup
results := make(chan int, len(jobs))
for _, j := range jobs {
    wg.Add(1)
    go func() {
        defer wg.Done()
        results <- work(j)
    }()
}
wg.Wait()
close(results)
```

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Warning: *A goroutine that blocks forever on a channel is a leak. Always have a way to stop it: close the channel or cancel a context.*

Context

```
ctx, cancel := context.WithTimeout(context.Background(), 2*time.Second)
defer cancel()
req, _ := http.NewRequestWithContext(ctx, http.MethodGet, url, nil)
select {
case <-ctx.Done():
    return ctx.Err()
case r := <-ch:
    _ = r
}
```

Pass `ctx` as the first parameter. Never store it in a struct.

HTTP service

```
mux := http.NewServeMux()
mux.HandleFunc("GET /healthz", func(w http.ResponseWriter, r *http.Request) {
    w.WriteHeader(http.StatusOK)
})
srv := &http.Server{Addr: ":8080", Handler: mux, ReadHeaderTimeout: 5 * time.Second}
go srv.ListenAndServe()
// on shutdown signal:
srv.Shutdown(ctx)
```

Warning: `http.ListenAndServe` *and the default client have no timeouts. Set them explicitly, or slow peers can hold connections forever.*

Testing

```
func TestSum(t *testing.T) {
    cases := []struct{ name string; a, b, want int }{
        {"zero", 0, 0, 0}, {"pos", 1, 2, 3},
    }
    for _, c := range cases {
        t.Run(c.name, func(t *testing.T) {
            if got := Sum(c.a, c.b); got != c.want { t.Fatalf("got %d want %d", got, c.want) }
        })
    }
}
```

```
go test -run TestSum -v ./pkg/...
go test -bench . -benchmem
go test -race ./...
```

Build for containers

```
CGO_ENABLED=0 GOOS=linux GOARCH=amd64 go build -trimpath -ldflags="-s -w" -o app ./cmd/api
```

```
FROM golang:1.23 AS build
WORKDIR /src
COPY go.mod go.sum ./
RUN go mod download
COPY . .
RUN CGO_ENABLED=0 go build -o /app ./cmd/api

FROM gcr.io/distroless/static-debian12
COPY --from=build /app /app
USER 65532:65532
ENTRYPOINT ["/app"]
```

Warning: *Pin the Go version in* `go.mod` *and in the build image, so local and CI builds use the same toolchain.*

Debugging

```
go test -race ./...
go tool pprof -http=:0 http://localhost:6060/debug/pprof/profile
curl -s localhost:6060/debug/pprof/goroutine?debug=1 | head -30
GODEBUG=gctrace=1 ./app
```

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Warning: *Never expose the pprof endpoint on a public port. Bind it to localhost or an internal network.*

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Helm and Kustomize

Charts, releases, values, upgrade and rollback commands, template functions and Kustomize overlays.

Repositories and search

```
helm repo add bitnami https://charts.bitnami.com/bitnami
helm repo update
helm repo list
helm search repo nginx --versions | head
helm show chart bitnami/nginx
helm show values bitnami/nginx > default-values.yaml
```

Install, upgrade, roll back

```
helm install web bitnami/nginx -n shop --create-namespace -f values-prod.yaml
helm upgrade --install web bitnami/nginx -n shop -f values-prod.yaml --atomic --timeout 5m
helm list -n shop
helm status web -n shop
helm history web -n shop
helm rollback web 2 -n shop
helm uninstall web -n shop
```

Keep overrides in values files, not `--set`. `--atomic` rolls back a failed upgrade.

Inspect and debug

```
helm template web bitnami/nginx -f values-prod.yaml
helm template web ./chart --debug
helm lint ./chart
helm get values web -n shop
helm get manifest web -n shop
helm diff upgrade web bitnami/nginx -n shop -f values-prod.yaml
```

`helm diff` needs the diff plugin. `helm template` never touches the cluster.

Chart layout

```
shop/
  Chart.yaml          name, version, appVersion
  values.yaml         defaults
  templates/
    deployment.yaml
    service.yaml
    _helpers.tpl      named templates
  charts/             dependencies
```

```
helm create shop
helm package ./shop
helm dependency update ./shop
helm push shop-0.1.0.tgz oci://registry.example.com/charts
```

Template syntax

```
{{ .Values.replicaCount }}                    read a value
{{ .Release.Name }}  {{ .Release.Namespace }} release info
{{ include "shop.fullname" . }}               call a helper
{{ .Values.image.tag | default "latest" | quote }}
{{- if .Values.ingress.enabled }} ... {{- end }}
{{- range .Values.hosts }} - {{ . }} {{- end }}
{{- toYaml .Values.resources | nindent 12 }}
```

`{{-` trims whitespace before the tag. Indentation errors are the most common bug.

Kustomize

```
kubectl kustomize overlays/production
kubectl apply -k overlays/production
kubectl diff -k overlays/production
kustomize build overlays/production | kubectl apply -f -
```

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
```

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```
  - ../../base
namespace: shop
images:
  - name: registry.example.com/shop/api
    newTag: 1.8.2
patches:
  - path: replica-patch.yaml
```

Base files stay valid YAML; overlays patch them per environment.

Find and inspect charts

```
helm repo add bitnami https://charts.bitnami.com/bitnami
helm repo update
helm search repo nginx --versions | head
helm show values bitnami/nginx > values-default.yaml
helm show chart bitnami/nginx
helm pull bitnami/nginx --untar --version 18.2.0
helm lint ./mychart
helm template web ./mychart -f values-prod.yaml | less
```

> **Warning:** *A chart is code that creates cluster resources. Read the templates and the values of third-party charts before installing, and pin the chart version with* `--version`*.*

Install, upgrade, roll back

```
helm install web ./mychart -n shop --create-namespace -f values-prod.yaml
helm upgrade --install web ./mychart -n shop -f values-prod.yaml --atomic --timeout 5m
helm upgrade web ./mychart -n shop --reuse-values --set image.tag=1.5
helm diff upgrade web ./mychart -n shop -f values-prod.yaml
helm list -A
helm status web -n shop
helm history web -n shop
helm rollback web 3 -n shop --wait
helm get values web -n shop --all
helm get manifest web -n shop
helm test web -n shop
```

> **Warning:** `--reuse-values` *keeps old values and ignores new chart defaults, which can hide a broken upgrade. Prefer passing the full values file each time.*

> **Warning:** `helm upgrade --install` *without* `--atomic` *can leave a half-upgraded release. Use* `--atomic` *and* `--timeout`*, and check* `helm history` *after a failure.*

> **Warning:** `helm diff` *is a separate plugin (*`helm plugin list`*). Install the helm-diff plugin before touching production, so you see the change before you apply it.*

Remove

```
helm uninstall web -n shop --dry-run
helm uninstall web -n shop
helm uninstall web -n shop --keep-history
```

> **Danger:** `helm uninstall` *deletes every resource of the release, and persistent volume claims may follow depending on the chart. Back up data first. CRDs installed from a chart's* `crds/` *folder are not removed by Helm.*

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Cloud security and IAM

Policies, roles, scopes, keys, private access, audit logs and an incident checklist for AWS and Azure.

Read an AWS policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject"],
    "Resource": "arn:aws:s3:::reports-prod/summaries/*"
  }]
}
```

Warning: *Action "" on Resource "" is administrator access. Never use it to make an error go away.*

Who am I and what can I do

```
aws sts get-caller-identity
aws iam list-attached-role-policies --role-name build
aws iam simulate-principal-policy --policy-source-arn ARN --action-names s3:PutObject --resource-a \
    rns ARN
aws iam generate-service-last-accessed-details --arn ARN
```

Warning: *Check which identity you are running as before changing any policy. Many denials come from the wrong role.*

Azure access

```
az account show
az role assignment list --assignee USER --all -o table
az role assignment create --assignee USER --role "Reader" --scope /subscriptions/SUB/resourceGroup \
    s/RG
az identity create --name app-identity --resource-group RG
```

Warning: *Assign roles at the lowest scope that works. A subscription-level Contributor can change everything below it.*

Secrets and keys

```
aws secretsmanager get-secret-value --secret-id prod/db --query SecretString --output text
aws kms describe-key --key-id alias/app-data
az keyvault secret show --vault-name kv-prod --name db-password --query value -o tsv
```

Warning: *Do not print secrets in logs or CI output. Fetch at run time, hold in memory, rotate with an overlap.*

Network exposure checks

```
aws s3api get-public-access-block --bucket NAME
aws ec2 describe-security-groups --filters Name=ip-permission.cidr,Values=0.0.0.0/0
az network nsg rule list --nsg-name NSG --resource-group RG -o table
```

Warning: *0.0.0.0/0 on SSH, RDP or a database port is an incident waiting to happen. Use session-based access instead.*

Audit trail lookups

```
aws cloudtrail lookup-events --lookup-attributes AttributeKey=EventName,AttributeValue=DeleteTrail \
     --max-results 5
aws cloudtrail get-trail-status --name org-trail
az monitor activity-log list --offset 1d -o table
```

Warning: *Alert on audit logging being stopped, root account use, new admin grants and public resources. Keep logs in a separate protected account.*

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Leaked credential order

```
1. Contain  - deactivate the key or revoke the session (do not delete yet)
2. Assess   - audit log: what did this credential do, from where
3. Remove   - attacker-created users, keys, servers, policies
4. Rotate   - related secrets; replace with a role
5. Learn    - how it leaked; add detection
```

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Kubernetes administration and operations

etcd backup, upgrades, node maintenance, RBAC, quotas, policies and a cluster debugging checklist.

etcd snapshot

```
ETCDCTL_API=3 etcdctl snapshot save /backup/etcd.db \
  --endpoints=https://127.0.0.1:2379 \
  --cacert=/etc/kubernetes/pki/etcd/ca.crt \
  --cert=/etc/kubernetes/pki/etcd/server.crt \
  --key=/etc/kubernetes/pki/etcd/server.key
etcdutl snapshot status /backup/etcd.db -w table
```

Warning: *Snapshots contain Secrets. Encrypt them, copy them off the node and test a restore on a schedule.*

Upgrade with kubeadm

```
kubeadm upgrade plan
kubeadm upgrade apply v1.NN.X
kubeadm upgrade node
kubeadm certs check-expiration
```

Warning: *Upgrade one minor version at a time, control plane first, and scan for removed APIs before you begin.*

Node maintenance

```
kubectl cordon node-1
kubectl drain node-1 --ignore-daemonsets --delete-emptydir-data
kubectl uncordon node-1
kubectl taint nodes gpu-1 gpu=true:NoSchedule
```

Warning: *A PodDisruptionBudget that allows zero disruptions makes drain hang. Check budgets before maintenance.*

RBAC

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: read-pods
  namespace: shop
subjects:
  - kind: Group
    name: shop-developers
    apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

Warning: *Do not bind cluster-admin to fix a permission error. Read the Forbidden message and grant the one verb on the one resource.*

Check permissions

```
kubectl auth can-i delete pods -n shop --as=jane
kubectl auth can-i --list -n shop --as=system:serviceaccount:shop:web
```

Quotas and autoscaling

```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: team-a
  namespace: team-a
spec:
  hard:
    requests.cpu: "20"
    requests.memory: 64Gi
    pods: "100"
```

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Warning: *The scheduler uses requests, not usage. Without realistic requests, autoscalers and quotas mislead you.*

Default deny network policy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny
  namespace: shop
spec:
  podSelector: {}
  policyTypes: ["Ingress", "Egress"]
```

Warning: *After default deny, allow DNS egress and the specific flows you need, and confirm your network plugin enforces policies.*

Debugging order

```
kubectl describe pod NAME
kubectl get nodes
kubectl describe node NODE
journalctl -u kubelet -n 100
kubectl get --raw=/readyz?verbose
```

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Continuous delivery with Argo CD

Applications, sync policies, hooks, projects, ApplicationSets and a debugging checklist.

An Application

```
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: shop-api
  namespace: argocd
spec:
  project: team-shop
  source:
    repoURL: https://example.com/platform/shop-config.git
    targetRevision: main
    path: apps/shop-api
  destination:
    server: https://kubernetes.default.svc
    namespace: shop
```

Warning: *The default project is unrestricted. Give each team its own project with limited sources and destinations.*

Everyday commands

```
argocd app list
argocd app get shop-api
argocd app diff shop-api
argocd app sync shop-api --dry-run
argocd app history shop-api
```

Warning: *Hand edits to live resources are reverted when self-heal is on. Change the repository instead.*

Automated sync

```
syncPolicy:
  automated:
    prune: true
    selfHeal: true
  syncOptions:
    - CreateNamespace=true
```

Warning: *Prune deletes whatever disappears from the source. A wrong path can look like delete everything. Protect stateful resources with Prune=false.*

Waves and hooks

```
metadata:
  annotations:
    argocd.argoproj.io/sync-wave: "-1"
    argocd.argoproj.io/hook: PreSync
    argocd.argoproj.io/hook-delete-policy: HookSucceeded
```

Warning: *Hooks can run more than once on retries. Make migrations idempotent.*

ApplicationSet

```
generators:
  - list:
      elements:
        - env: staging
        - env: production
template:
  metadata:
    name: "shop-{{env}}"
```

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Ignore expected differences

```
ignoreDifferences:
  - group: apps
    kind: Deployment
    jsonPointers:
      - /spec/replicas
```

Debug stages

```
1. Fetch   - repo access, revision, path
2. Render  - helm template / kustomize build with the same values
3. Apply   - permissions, project rules, webhooks, CRD order
4. Health  - ordinary Kubernetes debugging on the resources
```

Warning: *Never paste a secret into a values file. Reference an external secret store and rotate anything already committed.*

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LLM fundamentals

Tokens, context limits, API calls, sampling, cost, caching, evaluation and a debugging checklist for LLM features.

A first API call

```
import os
import anthropic

MODEL = os.environ["MODEL_ID"]
client = anthropic.Anthropic()          # reads ANTHROPIC_API_KEY
resp = client.messages.create(
    model=MODEL,
    max_tokens=500,
    system="You are a concise assistant.",
    messages=[{"role": "user", "content": "Explain a 502 error."}],
)
print(resp.content[0].text, resp.stop_reason, resp.usage)
```

> **Warning:** *Never put API keys in code, front-end JavaScript, logs or screenshots. Use a secret manager and rotate on exposure.*

Tokens and limits

```
if resp.stop_reason == "max_tokens":
    raise RuntimeError("reply was cut off")
print(resp.usage.input_tokens, resp.usage.output_tokens)
```

One token is about four English characters. Input, history, documents, tool results and the answer all count toward the context window.

> **Warning:** *A reply cut off by the length limit is incomplete. Truncated JSON will not parse, so always check the stop reason.*

Sampling

```
client.messages.create(model=MODEL, max_tokens=300, temperature=0.0,
                       messages=[{"role": "user", "content": "Classify: " + text}])
```

Low temperature for extraction, classification and code. Higher for brainstorming.

> **Warning:** *Temperature 0 does not guarantee identical output. Test properties and pass rates, not exact sentences.*

Errors and retries

```
import random, time

def with_retry(call, tries=4):
    for n in range(tries):
        try:
            return call()
        except anthropic.RateLimitError:
            time.sleep(2 ** n + random.random())
    raise RuntimeError("rate limited")
```

Retry 429 and 5xx with backoff and jitter. Do not retry 400, 401 or 403.

Cost estimate

```
def cost(inp, out, price_in, price_out):
    return inp / 1e6 * price_in + out / 1e6 * price_out
```

Output tokens usually cost more than input tokens. Use the current price list.

> **Warning:** *Agents and retry loops can spend a month of budget overnight. Set provider spend limits, alerts and a step cap.*

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Caching and streaming

```
with client.messages.stream(model=MODEL, max_tokens=500, messages=msgs) as s:
    for text in s.text_stream:
        print(text, end="", flush=True)
```

Put stable content first and changing content last to benefit from prompt caching.

Evaluate

```
{"id": "t-014", "input": "Card charged twice", "expect": {"category": "billing"}}
```

Keep 30 to 100 real cases, run every candidate several times, and compare pass rate, cost and latency. Re-run on any change.

Debug a bad answer

```
jq '{id, model, stop_reason, usage, error}' req.json
jq -r '.prompt' req.json | head -60
```

Warning: *Model output is untrusted. Validate it before it reaches a shell, a database or a customer.*

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Prompting and structured output

Prompt structure, examples, reasoning patterns, system prompts, JSON schemas, validation, retries and versioning.

Prompt template

```
You are helping {{audience}}.
Task: {{one clear instruction}}.
Rules:
- at most 3 sentences
- if the answer is not in <documents>, say "not stated"
<documents>{{material}}</documents>
<question>{{question}}</question>
Return JSON: {"answer": "...", "source": "..."}
```

Warning: *Replace adjectives with limits. "Be brief" varies; "at most three sentences" does not.*

Few-shot examples

```
<example>
<ticket>I was charged twice</ticket>
<answer>{"category": "billing"}</answer>
</example>
```

Use 3 to 5 varied, realistic examples with identical format, including a borderline case and an "unknown" case.

Warning: *Do not reuse your test cases as examples, and do not let examples contradict your rules. The model follows examples.*

Reasoning

```
Think step by step inside <thinking> tags.
Then give the final result in <answer> tags.
```

```
import re
m = re.search(r"<answer>(.*?)</answer>", reply, re.S)
answer = m.group(1).strip()
```

Warning: *Never show raw reasoning to users or treat an explanation as proof. Verify with code or a person.*

System prompt layout

```
<role>...</role>
<rules>...</rules>
<documents>...</documents>
<question>...</question>
```

Long documents first, question last. Keep tag names consistent.

Warning: *Do not put secrets in a system prompt. Assume users can make the model repeat it.*

JSON schema

```
{"type": "object",
 "properties": {"category": {"type": "string", "enum": ["billing", "account", "unknown"]}},
 "required": ["category"], "additionalProperties": false}
```

Methods: prompt only, forced tool use, schema-constrained output. Use enums and an explicit unknown value.

Validate and retry

```
for attempt in range(3):
    reply = call_model(prompt, feedback)
    try:
        data = json.loads(reply)
        validate(data, SCHEMA)
        return data
    except Exception as err:
        feedback = "Rejected: " + str(err)
raise RuntimeError("failed validation")
```

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> **Warning:** *Cap retries. A model that fails three times needs a changed prompt, not a fourth try. Never repair data by guessing.*

Version and test

```
prompts/triage/v7.txt     # changed: added unknown category
```

Run the evaluation set in CI, pin the model version, log the prompt version with each request, and release gradually.

Debug

```
python run_evals.py --case pii-014 --runs 10
```

Reproduce, reduce the prompt, change one thing, re-run the whole set, and add the case to the tests.

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Claude Skills

Skill folder layout, SKILL.md header, descriptions that trigger, structure, scripts, tests and a security review checklist.

Folder layout

```
incident-summary/
  SKILL.md
  template.md
  reference/format.md
  examples/outage-example.md
  scripts/check_summary.py
```

SKILL.md

```
---
name: incident-summary
description: Write a blameless incident summary from notes or a timeline. Use when the user asks f \
    or an incident summary, post-incident report or outage write-up.
---

Use template.md. Steps:
1. Collect start, end, impact, detection and mitigation.
2. Write "not stated" for anything the notes do not say.
3. Keep it under 300 words.
4. Run scripts/check_summary.py and fix what it reports.
```

> **Warning:** *The description decides whether the skill is ever used. Say what it does and when to use it, with real nouns and verbs. Check current name and length limits in the product documentation.*

Descriptions

```
Weak:   Helps with documents.
Better: Create and edit Word (.docx) files with headings, tables and tracked changes. Use when the \
     user asks for a Word document or to edit a .docx.
```

Add exclusions for neighbouring skills. Keep the key words first.

Progressive disclosure

```
Level 1: name and description (always)
Level 2: SKILL.md body (when triggered)
Level 3: linked files and scripts (when needed)
```

> **Warning:** *A giant SKILL.md wastes context. Move rarely needed detail to linked files, one level deep.*

Scripts

```
#!/usr/bin/env python3
"""Check a summary. Usage: python check_summary.py summary.md"""
import sys
text = open(sys.argv[1], encoding="utf-8").read()
problems = [] if len(text.split()) <= 300 else ["over 300 words"]
print("OK" if not problems else "\n".join(problems))
sys.exit(1 if problems else 0)
```

Use scripts for exact work. Keep output short and specific.

Test

```
Should trigger:     "write up last night's outage from these notes"
Should not trigger: "what does a 502 mean?"
```

Run 10 should-fire and 10 should-not requests, with and without the skill. Keep the task set in the repository.

Share safely

```
grep -rn "curl\|wget\|urlopen\|requests\.\|eval(\|exec(" skills/NAME/
```

> **Warning:** *A skill can steer Claude and include scripts that run. Read every file, look for network calls, downloads and shell commands, and run unknown skills in a sandbox first.*

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Debug

```
ls -la .claude/skills/NAME/
head -6 .claude/skills/NAME/SKILL.md
```

Check location, exact file name SKILL.md, header syntax, then the description, then the body, then scripts.

Warning: *Re-run your trigger and task tests after any model, product or description change.*

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Tool use and MCP

Tool definitions, the tool-use loop, MCP messages, a Python server, transports, approvals and injection defences.

Tool definition

```
{
  "name": "get_ticket",
  "description": "Fetch one support ticket by ID, such as TCK-1042. Returns title, status and prio \
    rity.",
  "input_schema": {
    "type": "object",
    "properties": {"ticket_id": {"type": "string"}},
    "required": ["ticket_id"]
  }
}
```

Warning: *Validate every tool call's arguments in your code. The model can send wrong types, extra fields or invalid values.*

The loop

```
for step in range(MAX_STEPS):
    reply = client.messages.create(model=MODEL, max_tokens=1000,
                                   tools=TOOLS, messages=messages)
    messages.append({"role": "assistant", "content": reply.content})
    if reply.stop_reason != "tool_use":
        break
    messages.append({"role": "user", "content": run_all_tools(reply)})
```

Return one tool result per call, matched by the call ID. Return errors as results.

Warning: *Always cap steps, tokens, time and cost, and stop on repeated identical calls.*

MCP messages

```
{"jsonrpc": "2.0", "id": 1, "method": "initialize"}
{"jsonrpc": "2.0", "id": 2, "method": "tools/list"}
{"jsonrpc": "2.0", "id": 3, "method": "tools/call",
 "params": {"name": "get_ticket", "arguments": {"ticket_id": "TCK-1042"}}}
```

Python server

```
from mcp.server.fastmcp import FastMCP

mcp = FastMCP("tickets")

@mcp.tool()
def get_ticket(ticket_id: str) -> dict:
    """Fetch one support ticket by ID. Returns title and status."""
    return TICKETS.get(ticket_id, {"error": "unknown ticket"})

if __name__ == "__main__":
    mcp.run()
```

Warning: *In stdio servers, write logs to standard error. Anything else on standard output breaks the protocol stream.*

Test and connect

```
npx @modelcontextprotocol/inspector python server.py
pytest tests/ -q
```

Use absolute paths and the virtual environment's Python in the host configuration.

Primitives

```
Tools      model-controlled actions
Resources  data by URI, attached by app or user
Prompts    templates the user chooses
```

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Security checklist

```
read-only credentials for read tools
approval for destructive, costly or external actions
no token passthrough; scoped short-lived tokens
tool output and fetched pages are untrusted data
```

> **Warning:** *Do not combine private data access, untrusted content and an outbound channel in one session without strong controls. Prompt wording is not a security boundary.*

Debug

```
python server.py < /dev/null | head -c 300
jq '.steps[] | {n, tool, input, is_error}' trace.json
```

Order: server alone, transport, protocol, definitions, model choices, loop.

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System design for platform and SRE

The design order, estimation numbers, building-block rules and failure-containment patterns on one page.

The order

```
1. Clarify    - users, what must never be wrong, SLO
2. Estimate   - requests per second, storage, bandwidth, peak
3. Sketch     - smallest design that meets the numbers
4. Deep-dive  - the risky part: data store, queue, rollout
5. Failures   - slow, down, wrong, overloaded, for every box
6. Trade-offs - say what you gave up
```

Numbers to remember

```
seconds per day     ~ 100,000 (86,400)
1M requests/day     ~ 10 per second
100M requests/day   ~ 1,000 per second
peak versus average ~ 3x unless known
memory read ~100 ns | SSD read ~100 us | same-region RTT ~1 ms | cross-continent RTT 100-200 ms
99.9% = 43 min/month | 99.99% = 4.3 min/month | 99.999% = 26 s/month
```

Building blocks in one line each

- **Cache:** jitter TTLs, coalesce refills, serve stale, warm before cutover, size the store for the cold case.
- **Queue:** assume at-least-once, make consumers idempotent, bound the queue, watch age not length.
- **Retry:** timeout shorter than the caller, exponential backoff with jitter, retry at one layer, use a budget.
- **Idempotency:** client key, stored with the result in one transaction.
- **Consistency:** pick per feature; fix stale reads with read-your-writes; alert on replication lag.
- **Quorum:** majority of an odd number; 3 tolerates 1, 5 tolerates 2; spread across zones.
- **Fencing:** the resource rejects a lower token; a lease alone is not enough.
- **Time:** monotonic for durations; never order by wall-clock across machines; handle negative differences.

Containing failure

```
failure domain   what one cause takes down together (process, host, zone, region, change)
cell             full independent copy for a slice of users, thin router, hard size limit
bulkhead         separate pools per dependency
shuffle shard    random worker subset per customer
canary           smallest stage with a stop signal and bake time
fail static      keep last known good on bad input
fail open/closed decide per check, write it down, test it
```

Overload

```
token bucket         steady refill, bounded burst
concurrency limit    cap in-flight work (protects slow paths)
shed by priority     batch first, critical last
signals              429 or 503 with Retry-After; clients back off with jitter
autoscaling          slow; limits hold the line meanwhile
```

Multi-region

```
RTO = time down       RPO = data lost, in time
backup/restore < pilot light < warm standby < active-active
survivors must carry the lost share (N regions need headroom)
two DNS providers; drill the failover; recovery tools off the failing system
```

Retry arithmetic

```
layers retrying  3 x 3 x 3  =  27 calls per user request at the bottom
fix: retry at one layer, jittered backoff, budget of ~10% of traffic
```

Warning: *Retries without jitter arrive in waves and keep a recovering service down.*

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Failover checks

```
# who is writable? exactly one should answer false for recovery
psql -h NODE -Atc "select pg_is_in_recovery()"
# replication lag in seconds on a replica
psql -Atc "select extract(epoch from now()-pg_last_xact_replay_timestamp())"
```

Warning: *Never promote a node without fencing the old primary first, or two writers will diverge.*

Data

```
partition key   many values, even load, keeps related data together; check the biggest value
range           cheap scans, hot end for sequential keys
hash            even load, scans hit all partitions
consistent hash ring + virtual nodes: adding a node moves ~1/N keys
storage engines B-tree: in-place pages, watch vacuum and bloat | LSM: append and compact, watch co \
    mpaction debt
replication     leader/follower | multi-leader (conflicts) | leaderless (R + W > N)
backup          RPO sets frequency, RTO sets restore speed; independent copy; timed restore test
```

Warning: *Replicas copy deletes. A backup you have not restored is a hope.*

By design: observability, security, cost

```
observability  SLI at the edge | golden signals | per-version and per-cell labels | trace id in lo \
    gs
               independent probe and paging path | dead-man's switch | cap cardinality
security       trust boundaries | least privilege | short-lived workload credentials | default-den \
    y network
               separate accounts and keys | audit logs | secret rotation runbook
cost/capacity  measure unit throughput at the SLO | N+1 and zone loss (3 zones: run under ~66%)
               list non-CPU limits | scale on leading signals | write down savings that cut redund \
    ancy
```

Change safety checklist

1. Can it be previewed on real data?
2. Smallest first stage and stop signal?
3. Previous good state and rollback time?
4. What does the consumer do with garbage?
5. Does the fixer depend on the broken thing?

Warning: *A second zone does not protect against a change pushed to every zone at once. Count shared things, not copies.*

Warning: *A status page, runbook or monitoring tool hosted on the failing platform will fail with it. Host them independently.*

Warning: *Automated quotas and limits need a floor. A limit that can reach zero is an outage waiting for a bad input.*

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MLOps

Checks, records and release steps that keep machine learning models reliable in production.

What to monitor

Layer	Signals	Why
Service	latency, errors, saturation	The usual health of the API
Inputs	null rate, range, new categories, distribution vs training	Catches broken or changed data
Predictions	average score, share per class	Early sign of drift
Quality	accuracy or error when labels arrive, vs a baseline	The real measure
Business	conversion, loss, complaints	What the model is for

Warning: *A healthy service does not mean a good model. Models fail quietly.*

Record for every training run

```
code version      a94f3c2
data version      churn-2026-03-01
parameters        learning_rate=0.05 max_depth=6
seed              42
environment       image digest, library versions, hardware
metrics           overall and per important group
artifacts         model file, evaluation report
```

Data checks before training

```
checks:
  - row_count: {min: 90000, max: 130000}
  - freshness: {column: event_date, max_age_hours: 30}
  - not_null: [customer_id, event_date]
  - range: {column: days_since_login, min: 0, max: 3650}
  - allowed_values: {column: plan_tier, values: [free, basic, pro, team]}
on_fail: stop_pipeline_and_page
```

Release steps

1. Register the model with run id, data version and metrics.
2. Package it in a pinned image and record the digest.
3. Gate: compare with production on a fixed set, including key groups.
4. Shadow on real traffic, then a small canary with quality checks.
5. Promote in steps. Keep the previous version ready.

Serving choices

Mode	Use when	Watch
Batch	results can be hours old	staleness, job failures
Online	the latest event matters	latency, timeouts, fallback
Streaming	high volume, near real time	backlog, duplicates

Drift alert example

```
- alert: FeatureDriftHigh
  expr: model_feature_psi{model="churn",feature="days_since_login"} > 0.25
  for: 6h
  labels: {severity: ticket}
```

Why it matters: *Drift is slow. A ticket with a window is better than a midnight page.*

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Debug a model bottom-up

1. What changed in the last week: data, feature code, model, image, traffic.
2. Inputs against training: null rate, range, new categories.
3. Served features against training features for the same record.
4. Replay a logged request with the same model version.
5. Only then the model, and last serving.

Warning: *Do not retrain before checking the data. A retrain on broken inputs repeats the fault.*

Warning: *Without logged inputs, output and model version, a bad prediction cannot be replayed.*

Warning: *Do not let a release pass on one overall metric. Check the groups that matter, such as new customers.*

Warning: *Do not retrain automatically with no gate. A retrain on bad data can produce a worse model.*

Commands for a model check

```
$ python check_inputs.py --window 48h --baseline training

$ python compare_features.py --sample 500 --serving-vs-training

$ python replay.py --request-id 7f3a --model churn-v14
```

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AIOps

Practical rules for anomaly detection, alert grouping, incident hints and safe auto-remediation.

Before you buy or build

- Write down a baseline: alerts per incident, time to acknowledge, time to find cause, false pages.
- Fix telemetry, service names and the change log first.
- Trial beside current practice for a few weeks. Keep it only if the numbers improve.

Anomaly detection ladder

Step	Method	Good for
1	Static threshold	Hard limits such as disk full
2	Rolling mean and spread	Metrics without a daily pattern
3	Seasonal baseline	Traffic, with daily and weekly rhythm
4	Forecast or model	Many series, after simple methods fail

```
sum(rate(http_requests_total{job="checkout"}[5m]))
/
sum(rate(http_requests_total{job="checkout"}[5m] offset 1w))
```

Reduce alert noise

```
route:
  group_by: [cluster, service]
  group_wait: 30s
inhibit_rules:
  - source_matchers: [alertname = "NodeDown"]
    target_matchers: [severity = "warning"]
    equal: [cluster, node]
```

Warning: *Silences need an expiry, a narrow matcher and a reason.*

Root cause hints

1. List changes just before the first symptom.
2. Overlay dependency health.
3. Rank by time gap, graph distance and size of anomaly.
4. Show evidence with every hint.
5. A person confirms before any production change.

Safe auto-remediation checklist

```
preconditions: [alert_still_firing, no_active_deploy, target_count_max: 2]
limits: {max_per_hour: 2, cooldown: 30m, kill_switch: automation/global}
verify: {after: 5m, expect: "queue_depth < 100"}
on_verify_fail: page_human
```

Why it matters: *Automation acts fast and at scale. Limits and a kill switch contain mistakes.*

Check an assistant's answer

1. Ask which evidence the claim rests on, and open it.
2. Turn the cause into a query you can run.
3. Line up alerts, deploys and changes on one timeline.
4. Record hits and misses.
5. A person approves every action.

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Warning: *A fluent answer is not evidence. A claim with no source is a guess.*

Warning: *Do not run a suggested rollback or restart before testing the claim against the data.*

Warning: *Do not give an AIOps tool write access before it has earned trust on read-only work.*

Warning: *Anomaly detection fires on harmless change too. Tune windows and keep the alert on user impact.*

Commands for checking a claim

```
$ promtool query instant http://prometheus:9090 'sum(rate(http_requests_total{code=~"5.."}[5m]))'

$ git log --since="2 hours ago" --oneline

$ kubectl rollout history deployment/checkout
```

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AIOps in practice

Approval tiers, action safety checklist, verification, kill switches, agent limits, injection defences and the evaluation steps to run before any AIOps feature acts.

Approval tiers

Tier	What it covers	Who acts
0	Read only: gather evidence, summarise, open a ticket	Automatic
1	Low risk and reversible: restart one pod, small capped scale-out	Automatic with logging
2	Moderate: rollback, drain, failover of a replica	Person approves with evidence
3	High risk: data, whole fleet, customer-visible region	Person does it

Warning: *Decide the tier when you write the action, not during the incident.*

Safe action template

```
action: restart-pod
tier: 1
preconditions:
  - service.healthy_replicas >= 2
  - no_active_deploy: true
scope: {max_pods_per_run: 1, max_runs_per_hour: 3}
timeout: 120s
verify: [pod.ready within 90s, error_ratio < 0.02 for 5m]
on_failure: page_oncall
```

Verify the symptom, not the command

```
sum(rate(http_requests_total{service="api",code=~"5.."}[1m]))
/ sum(rate(http_requests_total{service="api"}[1m])) < 0.02

max(db_connections_used / db_connections_max) < 0.85
```

Kill switch and limits

```
guardrails:
  global:
    kill_switch: flag://aiops/automation_enabled
    max_actions_per_hour: 20
  per_service:
    max_fleet_percent_per_run: 10
    require_human_if: [active_incident]
```

Check the switch before every action. Fail closed when the flag store cannot be read. Test the stop path every week.

Chaos experiment

```
apiVersion: chaos-mesh.org/v1alpha1
kind: PodChaos
metadata:
  name: api-pod-kill
spec:
  action: pod-kill
  mode: one
  selector:
    namespaces: [staging-api]
  duration: 60s
```

Write the hypothesis first: what should alert, what should run, how long recovery should take.

Agent safety checklist

- Narrow typed tools, no generic shell or free-form query

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- Read-only first, then proposals, then a human gate
- Step, time and token limits, with a full transcript
- Secrets outside the model context and egress allowlists
- Approval prompt shows the exact action, target and evidence

Evaluate before you trust

```
aiopsctl replay --suite incidents-2026 --detector mad-seasonal
aiopsctl shadow report --feature rca-ranker --weeks 4
aiopsctl eval topk --suite incidents-2026 --k 3
```

Report counts beside percentages. Add every wrong case to the regression suite.

Slow failures to monitor

```
sum by (action) (increase(remediation_rollbacks_total[7d]))
/ sum by (action) (increase(remediation_runs_total[7d]))
```

Watch rollback share, repeat fixes on one target, weekly precision and recall, and data freshness per source.

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Virtualization

virsh, qemu-img, bridges and migration commands, with the checks that explain a guest that will not start or a host that is slow.

Guests

```
virsh list --all
virsh start web-1
virsh shutdown web-1        # clean, asks the guest OS
virsh destroy web-1         # power off, like pulling the plug
virsh dominfo web-1
virsh console web-1
virsh domstate web-1 --reason
```

Warning: `destroy` *does not delete the guest, but it is an abrupt power-off. Use* `shutdown` *first.*

Memory and CPU

```
virsh nodememstats
virsh setmem web-1 4G --live
virsh setvcpus web-1 4 --live
virsh vcpuinfo web-1
free -m
grep -E "steal|st" <(top -bn1 | head -4)
```

Steal time above a few percent means the host CPU is oversubscribed.

Disks and snapshots

```
qemu-img info disk.qcow2
qemu-img create -f qcow2 -b base.qcow2 -F qcow2 overlay.qcow2
virsh pool-list
virsh pool-info default
virsh snapshot-list web-1
virsh snapshot-create-as web-1 before-upgrade
virsh snapshot-delete web-1 before-upgrade
```

Warning: *Never delete files in a pool by hand. Snapshot chains depend on their backing files.*

Network

```
virsh domiflist web-1
virsh net-list --all
ip link show type bridge
bridge link
bridge vlan show
```

Migration

```
virsh migrate --live web-1 qemu+ssh://host2/system
virsh domjobinfo web-1
virsh domjobabort web-1
```

If dirty rate is above bandwidth, the migration will not converge.

Quick diagnosis

Symptom	First look
Guest will not start	error text, free host memory, pool space
Guest paused (ioerror)	pool full or storage fault
Everything slow	host swap, steal time, disk latency
No network	`domiflist`, bridge exists, VLAN allowed on the switch

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Quick host checks

```
virsh list --all
virsh nodememstats
virsh domstats --cpu-total web-1
virsh pool-list --details
virsh domblklist web-1
```

Cautions

Danger: `virsh destroy` *powers a guest off instantly, like pulling the plug. Use* `virsh shutdown` *first.*

Warning: *Snapshots are not backups. They grow with every write and a full pool pauses all guests.*

Caution: *Overcommitting memory works until every guest is busy at once. Keep a reserve on the host.*

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Storage and filesystems

Disks, filesystems, LVM, RAID, network storage and the commands that explain a full, slow or missing volume.

See what is there

```
lsblk -f
blkid
df -hT
df -i
findmnt
mount | column -t
du -xh --max-depth=1 /var | sort -h | tail
```

Warning: `df` *and* `du` *can disagree. A deleted file that a process still holds open uses space until it closes.*

```
lsof +L1
```

Filesystems

```
mkfs.ext4 /dev/sdb1
mkfs.xfs /dev/sdb1
mount /dev/sdb1 /data
umount /data
tune2fs -l /dev/sdb1 | head
xfs_info /data
fsck -n /dev/sdb1     # check, no changes
```

fstab line:

```
UUID=7c1d...  /data  xfs  defaults,nofail  0  0
```

Test with `mount -a` before you reboot.

LVM

```
pvs; vgs; lvs
pvcreate /dev/sdc
vgextend vg0 /dev/sdc
lvextend -r -L +20G /dev/vg0/data   # -r grows the filesystem too
resize2fs /dev/vg0/data             # ext4
xfs_growfs /data                    # XFS, takes the mount point
```

XFS can grow but not shrink.

RAID

```
cat /proc/mdstat
mdadm --detail /dev/md0
mdadm /dev/md0 --add /dev/sdd1
```

Performance

```
iostat -x 1
iotop -o
vmstat 1
```

Watch `%util`, `await` and queue size, not only throughput.

Network storage

```
showmount -e server
mount -t nfs4 server:/export /mnt
nfsstat -m
iscsiadm -m session
ceph status
```

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Quick diagnosis

Symptom	First look
Full but nothing to delete	`df -i`, `lsof +L1`
Read-only filesystem	`dmesg`, `mount`, disk errors
Slow	`iostat -x`, queue and latency
Boot stops after mount change	`journalctl -xb`, fstab UUID, `nofail`

Cautions

Danger: `mkfs` *and* `wipefs` *destroy the data on a device. Check the name with* `lsblk` *and* `blkid` *every time.*

Warning: *Shrinking a filesystem or logical volume in the wrong order loses data. XFS cannot shrink at all.*

Warning: *A wrong line in /etc/fstab can stop boot. Use UUIDs and* `nofail`*, and run* `mount -a` *before rebooting.*

Caution: *RAID is not a backup. A deleted file is deleted on every disk.*

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KubeVirt

Install, VM objects, disks, networking, live migration, snapshots and a troubleshooting order for VMs on Kubernetes.

Check and install

```
ls -l /dev/kvm
kubectl apply -f kubevirt-operator.yaml
kubectl apply -f kubevirt-cr.yaml
kubectl -n kubevirt wait kv kubevirt --for condition=Available --timeout=10m
```

Warning: `useEmulation: true` *runs VMs in software and is for tests only. Real workloads need* `/dev/kvm`.

Day-to-day with virtctl

```
virtctl start web01
virtctl stop web01
virtctl console web01
virtctl vnc web01
virtctl ssh cloud-user@web01
virtctl expose vm web01 --port=80 --name=web01-svc
```

Warning: *Edit the VirtualMachine, not the VMI. Changes to a VMI vanish at the next start.*

Objects

Object	Meaning
VirtualMachine	Durable definition
VirtualMachineInstance	One running copy
DataVolume	Disk created and filled by CDI
VirtualMachineSnapshot	Point in time copy of disks and definition
MigrationPolicy	Rules for bandwidth and timeouts

Disks

```
apiVersion: cdi.kubevirt.io/v1beta1
kind: DataVolume
metadata:
  name: golden
spec:
  source:
    http:
      url: https://example.com/image.qcow2
  storage:
    resources:
      requests:
        storage: 20Gi
```

Danger: `containerDisk` *is ephemeral. Data written to it is lost when the VM restarts.*

Migration

```
virtctl migrate web01
kubectl get vmim
kubectl get vmi web01 -o jsonpath='{.status.conditions[?(@.type=="LiveMigratable")].message}'
```

Warning: *ReadWriteOnce disks and passthrough devices block live migration. Check LiveMigratable before a drain.*

Snapshots

```
kubectl apply -f snapshot.yaml
kubectl get vmsnapshot
```

Warning: *A snapshot on the same storage is not a backup. Copy data off the cluster with Velero or an export.*

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Troubleshooting order

Step	Look at
1	`kubectl get vm,vmi` phase and conditions
2	`kubectl describe vmi` and events
3	The virt-launcher pod: scheduling, OOMKilled
4	Node and virt-handler logs
5	Storage: PVC bound, DataVolume phase
6	Network: binding, Multus, Service selector
7	The guest: console, guest agent

Quick health sweep

```
kubectl get kubevirt -n kubevirt
kubectl get vmi -A
kubectl get dv,pvc -A
kubectl get vmim -A
virtctl console VMNAME
```

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Ceph

Daemons, status commands, pools, placement groups, cephadm, RBD, health warnings and a safe maintenance routine.

Status

```
ceph -s
ceph health detail
ceph osd tree
ceph osd df tree
ceph df
ceph pg stat
```

Warning: *Healthy placement groups are* `active+clean`*.* `inactive` *or* `incomplete` *means IO is blocked and is urgent.*

Daemons

Daemon	Job
MON	Cluster map and quorum, run 3 or 5
MGR	Metrics, dashboard, modules
OSD	Stores data on one disk
MDS	CephFS metadata
RGW	S3 and Swift gateway

Pools

```
ceph osd pool create vms 128
ceph osd pool set vms size 3
ceph osd pool set vms min_size 2
ceph osd pool application enable vms rbd
ceph osd pool autoscale-status
```

Danger: *Never set* `min_size 1`*. A pool can then accept writes that exist as a single copy.*

cephadm

```
cephadm bootstrap --mon-ip 10.0.0.11
ceph orch host add ceph2 10.0.0.12
ceph orch apply osd --all-available-devices
ceph orch ps
ceph orch upgrade start --ceph-version 18.2.4
```

Warning: *Zap a disk only after you confirm it is spare.* `ceph orch device zap` *destroys its data.*

RBD

```
rbd create rbd/db1 --size 100G
rbd du rbd/db1
rbd snap create rbd/db1@before-upgrade
rbd snap ls rbd/db1
```

Warning: *Old snapshots keep growing. Delete them on a schedule and watch* `rbd du`*.*

Maintenance

```
ceph osd set noout
ceph osd unset noout
ceph pg repair 7.3
```

Danger: *Reboot one host at a time and wait for HEALTH_OK. Several at once can drop PGs below min_size.*

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Capacity thresholds

Level	Meaning
85% nearfull	Warning, act now
90% backfillfull	Backfill to that OSD stops
95% full	Writes are blocked

Quick health sweep

```
ceph -s
ceph health detail
ceph osd tree
ceph df
ceph osd pool ls detail
```

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Vault

Seal and unseal, auth, policies, KV, dynamic secrets, PKI, transit and an incident response sequence.

Start and status

```
vault operator init -key-shares=5 -key-threshold=3
vault operator unseal
vault status
vault operator raft list-peers
```

Warning: *Dev mode (`vault server -dev`) is in memory with a known token. Never use it for real secrets.*

Auth

```
vault auth enable kubernetes
vault auth list
vault login -method=oidc
vault token lookup
vault token create -policy=app-read -ttl=15m
```

Warning: *Bind Kubernetes and cloud roles to an exact service account or role. Wildcards let any pod log in.*

Policy

```
path "secret/data/app/*" {
  capabilities = ["read"]
}
path "secret/metadata/app/*" {
  capabilities = ["list"]
}
```

Warning: *KV version 2 reads `secret/data/...`. A policy written without `data/` gives no access.*

KV

```
vault kv put secret/app/db user=app pass=example
vault kv patch secret/app/db pass=newexample
vault kv get -version=2 secret/app/db
vault kv destroy -versions=1 secret/app/db
```

Danger: `put` *replaces the whole secret. Use* `patch` *for a partial change.*

Dynamic and transit

```
vault read database/creds/orders-ro
vault lease revoke -prefix database/creds/orders-ro
vault write database/rotate-root/orders
vault write transit/encrypt/orders plaintext="$(echo -n secret | base64)"
```

Incident sequence

Step	Command or action
Identify	`vault token lookup -accessor "$ACCESSOR"`
Contain	`vault token revoke -accessor "$ACCESSOR"`
Sweep	`vault lease revoke -prefix <path>`
Investigate	Search the audit log by accessor
Rotate	Replace static secrets at the source

Danger: `vault operator seal` *stops every application. Use it only as a last resort.*

Quick health sweep

```
vault status
vault operator raft list-peers
vault audit list
vault token lookup
vault list sys/leases/lookup/database/creds/app
```

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SQL performance and database operations

Plans, indexes, locks, connections, replication and vacuum, with the queries that show what a busy PostgreSQL database is doing.

Find slow queries

```
SELECT calls, round(mean_exec_time) AS mean_ms, round(total_exec_time) AS total_ms, left(query, 80 \
    )
FROM pg_stat_statements ORDER BY total_exec_time DESC LIMIT 10;

EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM orders WHERE customer_id = 42;
```

Look for sequential scans on big tables, estimated rows far from actual rows, and sorts that spill to disk.

Indexes

```
CREATE INDEX CONCURRENTLY orders_cust_status ON orders (customer_id, status);
SELECT relname, indexrelname, idx_scan FROM pg_stat_user_indexes ORDER BY idx_scan LIMIT 10;
ANALYZE orders;
```

Warning: *A plain CREATE INDEX blocks writes. Use CONCURRENTLY on live tables.*

Sessions and locks

```
SELECT pid, state, now()-xact_start AS tx_age, wait_event_type, left(query, 60)
FROM pg_stat_activity WHERE state <> 'idle' ORDER BY tx_age DESC NULLS LAST;

SELECT pid, pg_blocking_pids(pid) AS blocked_by FROM pg_stat_activity WHERE cardinality(pg_blockin \
    g_pids(pid)) > 0;
SELECT pg_cancel_backend(2214);      -- cancel query
SELECT pg_terminate_backend(2214);   -- end session
```

Connections

```
SELECT state, count(*) FROM pg_stat_activity GROUP BY state;
SHOW max_connections;
```

Use a pooler. Set `idle_in_transaction_session_timeout`.

Vacuum and bloat

```
SELECT relname, n_live_tup, n_dead_tup, last_autovacuum FROM pg_stat_user_tables ORDER BY n_dead_t \
    up DESC LIMIT 10;
VACUUM (VERBOSE, ANALYZE) orders;
```

An old open transaction stops vacuum from removing dead rows.

Replication

```
SELECT now() - pg_last_xact_replay_timestamp() AS replica_lag;   -- on the replica
SELECT client_addr, state, sent_lsn, replay_lsn FROM pg_stat_replication;  -- on the primary
```

Quick diagnosis

Symptom	First look
One query slow	EXPLAIN ANALYZE, indexes, statistics
Everything slow	top statements, I/O, locks, connections
Too many connections	state grouping, idle in transaction, pooler
Table keeps growing	dead tuples, vacuum, old transactions
Replica behind	replay blocked, I/O, batch size

Cautions

Danger: `kill` *of a long transaction or* `pg_terminate_backend` *rolls it back. Check what it is doing first.*

Warning: *A plain* `CREATE INDEX` *blocks writes on the table. Use* `CREATE INDEX CONCURRENTLY` *on live systems.*

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Warning: *Raising* `max_connections` *does not fix a connection storm and can exhaust memory. Add a pooler.*

Caution: `VACUUM FULL` *locks the table and rewrites it. Use normal vacuum, or a repack tool, on busy tables.*

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GPU, AI infrastructure and LLM serving

Memory sums, nvidia-smi, vLLM flags, Kubernetes GPU scheduling and the metrics that explain slow or failing inference.

Memory sum

Item	Rule of thumb
Weights, 16-bit	parameters x 2 bytes (7B is about 14 GB)
Weights, 8-bit	parameters x 1 byte
Weights, 4-bit	parameters x 0.5 byte
KV cache	grows with context length x concurrent requests
Overhead	leave 10 to 20 percent free

Look at the GPU

```
nvidia-smi
nvidia-smi --query-gpu=name,memory.used,memory.total,utilization.gpu,temperature.gpu --format=csv
nvidia-smi dmon -c 5
nvidia-smi topo -m
```

> **Warning:** *High memory use with low utilisation usually means the GPU is waiting for the CPU or the network, not that it is busy.*

Serve with vLLM

```
vllm serve meta-llama/Llama-3.1-8B-Instruct \
  --max-model-len 8192 \
  --gpu-memory-utilization 0.90 \
  --tensor-parallel-size 1
curl localhost:8000/v1/models
curl localhost:8000/metrics | grep -E "num_requests|cache_usage"
```

Flag	Effect
`--max-model-len`	longest context, drives KV cache size
`--gpu-memory-utilization`	share of GPU memory the engine may use
`--tensor-parallel-size`	split the model over N GPUs

Kubernetes

```
resources:
  limits:
    nvidia.com/gpu: 1
tolerations:
- key: nvidia.com/gpu
  operator: Exists
  effect: NoSchedule

kubectl describe node gpu-1 | grep -A3 "nvidia.com/gpu"
kubectl get pods -A -o wide | grep -i nvidia
kubectl describe pod llm-0 | tail -15
```

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Signals to watch

Metric	Meaning
time to first token	how long users wait to see output
tokens per second	generation speed
requests waiting	demand above capacity
KV cache usage	near 100 percent means queueing and evictions
GPU memory and utilisation	fit and efficiency
Xid errors in dmesg	hardware or driver faults

Quick GPU checks

```
nvidia-smi --query-gpu=name,memory.used,memory.total,utilization.gpu,temperature.gpu --format=csv
nvidia-smi dmon -c 3
kubectl describe node gpu-1 | grep -A3 nvidia.com/gpu
kubectl logs deploy/llm --tail=50
```

Test a server

```
curl -s localhost:8000/v1/models
curl -s localhost:8000/v1/chat/completions -H "Content-Type: application/json" -d '{"model":"m","m \
    essages":[{"role":"user","content":"hi"}],"max_tokens":16}'
curl -s localhost:8000/metrics | grep num_requests
```

Cautions

Danger: *Setting* `--gpu-memory-utilization` *near 1.0 leaves no room for spikes and invites out-of-memory crashes.*

Warning: `nvidia-smi --gpu-reset` *and driver changes interrupt every process on the card. Drain the node first.*

Warning: *Autoscaling a model server on CPU use does not work, as the work happens on the GPU. Use queue depth or latency.*

Caution: *A long context grows the KV cache for every request. Set a maximum that fits your memory.*

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OpenStack and private cloud

The openstack client commands and checks for identity, compute, networking, volumes and troubleshooting a private cloud.

Sign in and check the basics

```
env | grep ^OS_
openstack token issue
openstack catalog list
openstack project list
openstack quota show
```

A working `token issue` proves Keystone and your credentials are good.

Compute

```
openstack server list
openstack server show app-9 -c status -c fault
openstack server create --flavor m1.small --image ubuntu --network private app-9
openstack server reboot app-9
openstack server delete app-9
openstack flavor list
openstack image list
openstack console log show app-9
```

Warning: *Servers in ERROR still count against your quota. Delete them.*

Scheduling and health

```
openstack compute service list
openstack hypervisor list
openstack hypervisor stats show
openstack resource provider usage show PROVIDER_ID
```

Capacity is per node. A big flavor can fail even when the total free memory is large.

Networking

```
openstack network list
openstack subnet list
openstack port list --server app-9
openstack security group list
openstack security group rule list web --long
openstack security group rule create --proto tcp --dst-port 22 --remote-ip 203.0.113.7/32 web
openstack floating ip create public
openstack server add floating ip app-9 203.0.113.45
```

Volumes

```
openstack volume list
openstack volume create --size 20 data
openstack server add volume app-9 data
openstack volume service list
```

Troubleshooting

Symptom	First look
HTTP 401	credentials, domain names, token expiry, clock
HTTP 403	role and policy, or quota
ERROR, no valid host	service state, per-node free resources, scheduler log
Cannot reach server	console log, security group, floating IP, router
Many services fail at once	RabbitMQ, database, disk space
Stuck state	compare hypervisor with database before reset

Follow one request through logs with its request id (`req-...`).

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Quick health checks

```
openstack compute service list
openstack network agent list
openstack volume service list
openstack hypervisor stats show
openstack server show NAME -c status -c fault
```

Cautions

Danger: `openstack server delete` *removes the server and its ephemeral disk. Check for volumes and snapshots you need first.*

Warning: *A security group rule open to 0.0.0.0/0 on every port exposes every server that uses the group.*

Warning: *Evacuate a node only after it is fenced or powered off, or two copies of a server can run at once.*

Caution: *Back up the database before any upgrade. Many migrations cannot be reversed.*

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Bare metal and datacenter

Remote management, provisioning, disks and RAID, bonding and the first checks on a sick server.

Talk to the BMC

```
ipmitool -H BMC_IP -U USER -P PASS chassis power status
ipmitool -H BMC_IP -U USER -P PASS chassis power cycle
ipmitool -H BMC_IP -U USER -P PASS sel elist
ipmitool -H BMC_IP -U USER -P PASS sensor list
```

Warning: `power cycle` *and* `power off` *cut power at once. Drain the host first unless it is already hung.*

Check disks

```
lsblk -f
sudo smartctl -H /dev/sda
sudo smartctl -a /dev/sda | grep -E "Reallocated|Pending|Power_On"
cat /proc/mdstat
sudo mdadm --detail /dev/md0
```

Replace a failed RAID member

```
sudo mdadm /dev/md0 --fail /dev/sdb
sudo mdadm /dev/md0 --remove /dev/sdb
sudo mdadm /dev/md0 --add /dev/sdc
watch cat /proc/mdstat
```

Network and bonding

```
cat /proc/net/bonding/bond0
ip -s link show bond0
ethtool eno1 | grep -E "Speed|Link detected"
dmesg | grep -i bond
```

PXE problems

```
sudo journalctl -u dnsmasq -f
grep -E "dhcp-boot|next-server|filename" /etc/dnsmasq.conf
sudo tcpdump -ni eth0 port 67 or port 68 or port 69
```

A sick server checklist

1. Drain the workloads.
2. Read the BMC event log, SMART, kernel log and switch counters.
3. Decide which part is at fault. Do not swap parts at random.
4. Change one thing, test, record it by asset tag.

Quick inventory

```
lscpu | head -5
free -h
lsblk -o NAME,SIZE,MODEL,SERIAL
dmidecode -s system-serial-number
ip -br link
```

Cautions

Danger: `ipmitool chassis power off` *and* `power cycle` *cut power with no clean shutdown. Drain the workload first, and check you have the right BMC address.*

Warning: *Never run* `mdadm --zero-superblock` *or* `--create` *on a disk that still holds data. Check* `mdadm --detail` *and* `lsblk` *first.*

Warning: *A BMC on a shared or routable network is an entry point to the machine. Keep it on an isolated management network and change default passwords.*

Caution: *Replacing a failed disk starts a rebuild that reads every other disk. Confirm the backup works before you pull the disk.*

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Service mesh and Kubernetes networking

Check Services and endpoints, DNS, network policy, sidecars, sync and the response flags in proxy logs.

Service and endpoints

```
kubectl get svc,endpoints web -n shop
kubectl get svc web -n shop -o jsonpath='{.spec.selector}'
kubectl get pods -n shop --show-labels
kubectl get endpointslices -n shop
```

DNS from a pod

```
kubectl exec -n shop deploy/web -- cat /etc/resolv.conf
kubectl exec -n shop deploy/web -- nslookup db.shop
kubectl logs -n kube-system -l k8s-app=kube-dns --tail=20
```

Network policy

```
kubectl get networkpolicy -n shop
kubectl describe networkpolicy default-deny -n shop
kubectl exec -n shop deploy/web -- curl -s -m 2 http://db:5432
```

Tip: *A timeout where you expect a refusal often means a policy is dropping packets.*

Mesh health

```
istioctl version
istioctl proxy-status
istioctl analyze -n shop
kubectl get pods -n shop
kubectl label namespace shop istio-injection=enabled
kubectl rollout restart deployment -n shop
```

Proxy logs and flags

```
kubectl logs web-7d -c istio-proxy --tail=20
```

Flag	Meaning
UF	Upstream connection failure
UH	No healthy upstream
NR	No route configured
UC	Upstream connection terminated
URX	Retry limit exceeded

Method for a failing call

1. Find the first failing hop.
2. `istioctl proxy-status`: is the config SYNCED?
3. `istioctl analyze`: is the config valid?
4. Read the response flag.
5. Check endpoints, then policies (PeerAuthentication, AuthorizationPolicy).
6. Call the pod IP directly to take the mesh out of the picture.

Quick mesh checks

```
istioctl analyze -n shop
istioctl proxy-status
istioctl proxy-config clusters POD.shop | head
kubectl get peerauthentication,authorizationpolicy -A
```

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Cautions

Danger: *Switching a namespace to STRICT mTLS rejects every client without a sidecar. Use PERMISSIVE first and check what is still plaintext.*

Warning: *Retries at several layers multiply load. Keep one layer of retries and set a budget, or a slowdown becomes an outage.*

Warning: *Weights in a VirtualService must add up to 100, and every subset must exist in a DestinationRule, or requests fail with 503.*

Caution: *A default-deny network policy blocks DNS too. Allow egress to kube-dns or every lookup will fail.*

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Reverse proxies and load balancers

nginx and HAProxy basics, the status codes you will see, timeouts, health checks and a debugging order for 502 and 504.

Test and reload safely

```
nginx -t                      # syntax and file check, never reload without it
nginx -T | less               # print the full effective configuration
nginx -s reload               # graceful reload, old workers finish their requests
tail -f /var/log/nginx/error.log
```

A minimal reverse proxy

```
upstream app {
    server 10.0.2.14:8080 max_fails=3 fail_timeout=10s;
    server 10.0.2.15:8080;
    keepalive 32;
}
server {
    listen 443 ssl;
    server_name shop.example.com;
    location / {
        proxy_pass http://app;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $remote_addr;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_connect_timeout 3s;
        proxy_read_timeout 30s;
    }
}
```

Status codes from the proxy

Code	Meaning	First check
502	No valid answer from the upstream	Error log: connection refused or reset
503	No healthy upstream or limit reached	Health checks, connection limits
504	Upstream too slow	Backend latency, database, timeouts
429	Rate limit applied	Limit settings and client behaviour

Look from outside and inside

```
curl -sI https://shop.example.com | head -5
curl -s -o /dev/null -w '%{http_code} %{time_connect} %{time_starttransfer}\n' https://shop.exampl \
    e.com/
ss -ltnp | grep -E ':80|:443|:8080'
openssl s_client -connect shop.example.com:443 -servername shop.example.com </dev/null | openssl x \
    509 -noout -dates
```

HAProxy essentials

```
haproxy -c -f /etc/haproxy/haproxy.cfg     # validate configuration
echo "show stat" | socat stdio /run/haproxy/admin.sock | cut -d, -f1,2,18
```

Debugging order

1. Read the error log line and note the upstream address and error text.
2. Connect to the backend from the proxy host with curl.
3. Check backend health, CPU, database and connection pools.
4. Review the last change: deploy, configuration, certificate.
5. Turn off risky retries, add limits, fix the slow dependency, then tune timeouts.

Safe changes

```
nginx -t && nginx -s reload
```

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```
curl -sv -H 'Host: app.example.com' http://127.0.0.1/ -o /dev/null

curl -sI https://app.example.com | grep -i 'strict-transport\|server'

haproxy -c -f /etc/haproxy/haproxy.cfg
```

Warning: *Always run* `nginx -t` *before a reload. A bad config makes the reload fail, and a restart instead of a reload drops live connections.*

Warning: *A trailing slash in* `proxy_pass` *changes the path sent to the backend. Test one URL by hand after every change.*

Danger: *Never cache responses that carry cookies or* `Authorization` *headers without a key that separates users. One user can see another user's page.*

Warning: *Retrying non-idempotent requests (POST) on a timeout can charge or create twice. Limit* `proxy_next_upstream` *to safe methods.*

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Messaging and streaming

RabbitMQ and Kafka commands for queues, topics, consumer groups and lag, with the delivery rules to remember.

Delivery rules

Term	Rule to remember
At-least-once	Duplicates are possible, so handlers must be idempotent
Ordering	Guaranteed only within one queue or partition
Acknowledge	After the work is saved, not before
Dead letters	Every queue needs a failed-message path and an alert on it

RabbitMQ

```
rabbitmqctl list_queues name messages_ready messages_unacknowledged consumers
rabbitmqctl list_bindings
rabbitmqctl list_connections name state
rabbitmqadmin declare queue name=billing durable=true
rabbitmqadmin publish exchange=events routing_key=order.paid.eu payload='{"id":4411}'
```

Kafka topics

```
kafka-topics.sh --bootstrap-server b1:9092 --list
kafka-topics.sh --bootstrap-server b1:9092 --describe --topic orders
kafka-topics.sh --bootstrap-server b1:9092 --describe --under-replicated-partitions
kafka-configs.sh --bootstrap-server b1:9092 --entity-type topics --entity-name orders --describe
```

Kafka consumers and lag

```
kafka-consumer-groups.sh --bootstrap-server b1:9092 --list
kafka-consumer-groups.sh --bootstrap-server b1:9092 --describe --group invoicer
kafka-console-consumer.sh --bootstrap-server b1:9092 --topic orders --from-beginning --max-message \
    s 5
```

Producer and consumer settings that matter

Setting	Safe starting value
acks	all
enable.idempotence	true
min.insync.replicas	2 with replication factor 3
enable.auto.commit	false
max.poll.interval.ms	longer than your slowest batch

Debugging order

1. Are consumers running and connected?
2. Is lag on one partition or all of them?
3. Is a dependency slow, or is a message failing in a loop?
4. Is the broker healthy: disk, under-replicated partitions, controller?
5. Trace one message id from producer to database.

Daily checks

```
kafka-consumer-groups.sh --bootstrap-server $B --describe --group orders

kafka-topics.sh --bootstrap-server $B --describe --topic orders

rabbitmqctl list_queues name messages consumers

kafka-console-consumer.sh --bootstrap-server $B --topic orders --from-beginning --max-messages 5
```

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Warning: *Resetting consumer group offsets skips or replays messages. Stop the group first and use* `--dry-run` *before* `--execute`*.*

Danger: *Deleting a topic or purging a queue cannot be undone. Check the name twice and confirm the environment.*

Warning: *Auto-acknowledge loses messages when a consumer crashes mid-work. Acknowledge after the work is stored.*

Warning: *Lowering retention deletes old data on the next cleanup. Check what slow consumers still need first.*

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CloudFormation and Bicep

Commands to validate, preview, deploy and debug CloudFormation stacks and Bicep deployments, and the checks to run first.

CloudFormation

```
aws cloudformation validate-template --template-body file://app.yaml
cfn-lint app.yaml
aws cloudformation deploy --template-file app.yaml --stack-name app-dev --parameter-overrides Env= \
    dev
aws cloudformation describe-stack-events --stack-name app-dev --max-items 20
aws cloudformation describe-stacks --stack-name app-dev --query 'Stacks[0].StackStatus'
```

Preview an update

```
aws cloudformation create-change-set --stack-name app --change-set-name c1 --template-body file:// \
    app.yaml
aws cloudformation describe-change-set --stack-name app --change-set-name c1 \
  --query 'Changes[].ResourceChange.[Action,LogicalResourceId,Replacement]'
aws cloudformation execute-change-set --stack-name app --change-set-name c1
```

Drift

```
aws cloudformation detect-stack-drift --stack-name app
aws cloudformation describe-stack-resource-drifts --stack-name app --stack-resource-drift-status-f \
    ilters MODIFIED DELETED
```

Bicep

```
az bicep build --file main.bicep
az deployment group what-if --resource-group rg-app --template-file main.bicep --parameters env=de \
    v
az deployment group create --resource-group rg-app --template-file main.bicep --parameters env=dev
az deployment operation group list --resource-group rg-app --name main
```

Stack states

Status	What to do
ROLLBACK_COMPLETE	Read the first failed event, delete the stack, deploy again
UPDATE_ROLLBACK_FAILED	Fix the cause, then continue the rollback
DELETE_FAILED	Empty or retain the blocking resource, then delete

Safe habits

1. Run lint, policy checks and a preview on every pull request.
2. Read Replacement and Delete lines before approving.
3. Protect stateful resources with snapshot or retain policies.
4. Deploy from a pipeline with a federated identity, not a stored key.
5. Check drift on a schedule and fix it through code.

Safe deploys

```
aws cloudformation create-change-set --stack-name app --change-set-name c1 --template-body file:// \
    t.yaml

aws cloudformation describe-stack-events --stack-name app --max-items 10

az deployment group what-if -g rg-app -f main.bicep

cfn-lint t.yaml
```

Danger: *A property change can replace a resource, and replacing a database deletes its data. Read the change set* `Replacement` *column before you execute it.*

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Warning: *Set* `DeletionPolicy: Retain` *or* `Snapshot` *on stateful resources before you delete or rename them.*

Warning: *Bicep complete mode deletes resources in the group that are not in the file. Run* `what-if` *first.*

Warning: *Editing resources by hand causes drift. The next update can undo your change or fail.*

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Google Cloud

gcloud commands for projects, networks, compute, storage, Cloud Run, GKE and logs, plus a debugging order for permission errors.

Configuration

```
gcloud auth login
gcloud config configurations list
gcloud config set project acme-pay-dev
gcloud services enable compute.googleapis.com run.googleapis.com
gcloud projects get-iam-policy acme-pay-dev --format=json
```

Network and compute

```
gcloud compute networks create prod-vpc --subnet-mode=custom
gcloud compute firewall-rules list --filter="network:prod-vpc"
gcloud compute instances list --filter="labels.env=prod"
gcloud compute ssh web-1 --zone europe-west1-b --tunnel-through-iap
gcloud compute instance-groups managed list-instances web --region europe-west1
```

Storage and data

```
gcloud storage ls --long gs://acme-exports/
gcloud storage buckets describe gs://acme-exports --format='value(iamConfiguration)'
gcloud sql instances list
bq query --dry_run --use_legacy_sql=false 'select count(*) from logs.events where dt = "2026-11-05 \
    "'
```

Containers

```
gcloud run deploy web --image=REGION-docker.pkg.dev/PROJECT/app/web:1.0 --region=europe-west1 --no \
    -allow-unauthenticated
gcloud run services describe web --region europe-west1
gcloud container clusters get-credentials prod --region europe-west1
kubectl get pods -A
```

Logs

```
gcloud logging read 'resource.type="cloud_run_revision" AND severity>=ERROR' --limit=20 --freshnes \
    s=1h
gcloud pubsub subscriptions describe billing
```

Common errors

Message	First step
PERMISSION_DENIED	Policy Troubleshooter, then a narrow role
API has not been used	Enable the API in the project
Quota exceeded	Check quotas, request an increase or change region
Blocked by organization policy	Read the constraint, use the approved path
VPC Service Controls violation	Read the audit log violation entry

Safe habits

```
gcloud config list

gcloud projects get-iam-policy my-project --format=json

gcloud compute instances list --filter="status=RUNNING"

gcloud logging read 'severity>=ERROR' --limit=10 --freshness=1h

gcloud run services describe web --region=europe-west1
```

Warning: *Check* `gcloud config list` *before every change. The active project decides where a command runs.*

Danger: *Granting* `roles/owner` *or* `roles/editor` *to a service account gives it broad power. Use the narrowest predefined role.*

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Warning: *Making a bucket public with* `allUsers` *exposes every object. Use uniform access and signed URLs instead.*

Warning: *Deleting a project starts a 30 day shutdown for everything in it. Confirm the project ID first.*

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